

Supplementary Material: NetShield

Nuonan Ouyang, Adrian Shatte, Zhigang Lu, Chao Chen, and Wei Xiang

I. Evidence Classification and Scope

The formal Raspberry Pi campaign contains 12 sessions and 216 accepted arms. Task C reconstructs timing from accepted formal hardware timestamps. Task A is offline replay, Task B is a software sensitivity, and Task D is discrete-event conformance. None is a new Raspberry Pi experiment. This supplementary build does not modify the frozen formal raw evidence.

TABLE I
Evidence classes.

Component	Evidence type	Permitted interpretation
Formal campaign	Hardware	Three-Pi measurements
Full-raw audit	Raw reconstruction	Accepted-arm traceability
Task A	Offline replay	Frozen-selector sensitivity
Task B	Software sensitivity	Phase-desynchronization counterfactual
Task C	Raw reconstruction	Accepted-hardware timestamp metrics
Task D	Discrete-event conformance	Analytical boundary check

II. Frozen Action Table and I-QPG Thresholds

TABLE II
Frozen action lattice and $v = 1$ thresholds against H-NONE.

Action	Payload (B)	Utility	Threshold Q (B)
L-NONE	0	0.758501	–
L-ALERT	256	0.825168	221.84
L-SUMMARY	4096	0.891835	30.14
M-NONE	0	0.766968	–
M-ALERT	256	0.833635	254.91
M-SUMMARY	8192	0.900302	16.10
H-NONE	0	0.768378	–
H-ALERT	256	0.835044	260.41
H-SUMMARY	16384	0.901711	8.14
H-FULL	131072	0.968378	1.53

III. Theorem Derivations

For m remotely represented probes and k full-evidence probes, payload is $mS + k(B - S)$. Hence one-round feasibility is

$$kB + (m - k)S \leq C,$$

and, for $C \geq mS$,

$$k^*(C, m) = \min \left\{ m, \left\lfloor \frac{C - mS}{B - S} \right\rfloor \right\}.$$

With $\rho = C/(nB)$, $\sigma = S/B$, $r = m/n$, and $f = k/n$, the continuous frontier is $\sigma r + (1 - \sigma)f \leq \rho$. For synchronized fixed-full overload, conservation gives $Q_\Sigma(T) = T(nB - C)$. At mid capacity the frozen I-QPG state alternates between H-FULL and H-NONE because the post-full queue exceeds every positive-payload threshold and the zero-payload round drains the residual.

IV. Complete V-Sweep Results

The $v = 1$ identity gate covers 36 accepted D cells and four deterministic checks per cell. Low-capacity cycle/action composition changes with v ; no all- v cycle-identity claim is made.

TABLE III: Complete frozen-selector v -sweep results (432 retained rows).

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S01	1	low	0.244444	9191424	87382	4
F12_S02	1	low	0.244444	9191424	87382	4
F12_S03	1	low	0.244444	9191424	87382	4
F12_S04	1	low	0.244444	9191424	87382	4
F12_S05	1	low	0.244444	9191424	87382	4
F12_S06	1	low	0.244444	9191424	87382	4
F12_S07	1	low	0.244444	9191424	87382	4
F12_S08	1	low	0.244444	9191424	87382	4
F12_S09	1	low	0.244444	9191424	87382	4
F12_S10	1	low	0.244444	9191424	87382	4
F12_S11	1	low	0.244444	9191424	87382	4
F12_S12	1	low	0.244444	9191424	87382	4
F12_S01	1	mid	0.500000	17694720	43691	2
F12_S02	1	mid	0.500000	17694720	43691	2
F12_S03	1	mid	0.500000	17694720	43691	2
F12_S04	1	mid	0.500000	17694720	43691	2
F12_S05	1	mid	0.500000	17694720	43691	2
F12_S06	1	mid	0.500000	17694720	43691	2
F12_S07	1	mid	0.500000	17694720	43691	2
F12_S08	1	mid	0.500000	17694720	43691	2
F12_S09	1	mid	0.500000	17694720	43691	2
F12_S10	1	mid	0.500000	17694720	43691	2
F12_S11	1	mid	0.500000	17694720	43691	2
F12_S12	1	mid	0.500000	17694720	43691	2
F12_S01	1	high	1.000000	35389440	0	1
F12_S02	1	high	1.000000	35389440	0	1
F12_S03	1	high	1.000000	35389440	0	1
F12_S04	1	high	1.000000	35389440	0	1
F12_S05	1	high	1.000000	35389440	0	1
F12_S06	1	high	1.000000	35389440	0	1
F12_S07	1	high	1.000000	35389440	0	1
F12_S08	1	high	1.000000	35389440	0	1
F12_S09	1	high	1.000000	35389440	0	1
F12_S10	1	high	1.000000	35389440	0	1
F12_S11	1	high	1.000000	35389440	0	1
F12_S12	1	high	1.000000	35389440	0	1
F12_S01	3	low	0.244444	9191424	87382	4
F12_S02	3	low	0.244444	9191424	87382	4
F12_S03	3	low	0.244444	9191424	87382	4
F12_S04	3	low	0.244444	9191424	87382	4
F12_S05	3	low	0.244444	9191424	87382	4
F12_S06	3	low	0.244444	9191424	87382	4
F12_S07	3	low	0.244444	9191424	87382	4
F12_S08	3	low	0.244444	9191424	87382	4
F12_S09	3	low	0.244444	9191424	87382	4
F12_S10	3	low	0.244444	9191424	87382	4
F12_S11	3	low	0.244444	9191424	87382	4
F12_S12	3	low	0.244444	9191424	87382	4
F12_S01	3	mid	0.500000	17694720	43691	2
F12_S02	3	mid	0.500000	17694720	43691	2
F12_S03	3	mid	0.500000	17694720	43691	2
F12_S04	3	mid	0.500000	17694720	43691	2
F12_S05	3	mid	0.500000	17694720	43691	2
F12_S06	3	mid	0.500000	17694720	43691	2

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S07	3	mid	0.500000	17694720	43691	2
F12_S08	3	mid	0.500000	17694720	43691	2
F12_S09	3	mid	0.500000	17694720	43691	2
F12_S10	3	mid	0.500000	17694720	43691	2
F12_S11	3	mid	0.500000	17694720	43691	2
F12_S12	3	mid	0.500000	17694720	43691	2
F12_S01	3	high	1.000000	35389440	0	1
F12_S02	3	high	1.000000	35389440	0	1
F12_S03	3	high	1.000000	35389440	0	1
F12_S04	3	high	1.000000	35389440	0	1
F12_S05	3	high	1.000000	35389440	0	1
F12_S06	3	high	1.000000	35389440	0	1
F12_S07	3	high	1.000000	35389440	0	1
F12_S08	3	high	1.000000	35389440	0	1
F12_S09	3	high	1.000000	35389440	0	1
F12_S10	3	high	1.000000	35389440	0	1
F12_S11	3	high	1.000000	35389440	0	1
F12_S12	3	high	1.000000	35389440	0	1
F12_S01	10	low	0.300000	10838016	87386	10
F12_S02	10	low	0.300000	10838016	87386	10
F12_S03	10	low	0.300000	10838016	87386	10
F12_S04	10	low	0.300000	10838016	87386	10
F12_S05	10	low	0.300000	10838016	87386	10
F12_S06	10	low	0.300000	10838016	87386	10
F12_S07	10	low	0.300000	10838016	87386	10
F12_S08	10	low	0.300000	10838016	87386	10
F12_S09	10	low	0.300000	10838016	87386	10
F12_S10	10	low	0.300000	10838016	87386	10
F12_S11	10	low	0.300000	10838016	87386	10
F12_S12	10	low	0.300000	10838016	87386	10
F12_S01	10	mid	0.500000	17694720	43691	2
F12_S02	10	mid	0.500000	17694720	43691	2
F12_S03	10	mid	0.500000	17694720	43691	2
F12_S04	10	mid	0.500000	17694720	43691	2
F12_S05	10	mid	0.500000	17694720	43691	2
F12_S06	10	mid	0.500000	17694720	43691	2
F12_S07	10	mid	0.500000	17694720	43691	2
F12_S08	10	mid	0.500000	17694720	43691	2
F12_S09	10	mid	0.500000	17694720	43691	2
F12_S10	10	mid	0.500000	17694720	43691	2
F12_S11	10	mid	0.500000	17694720	43691	2
F12_S12	10	mid	0.500000	17694720	43691	2
F12_S01	10	high	1.000000	35389440	0	1
F12_S02	10	high	1.000000	35389440	0	1
F12_S03	10	high	1.000000	35389440	0	1
F12_S04	10	high	1.000000	35389440	0	1
F12_S05	10	high	1.000000	35389440	0	1
F12_S06	10	high	1.000000	35389440	0	1
F12_S07	10	high	1.000000	35389440	0	1
F12_S08	10	high	1.000000	35389440	0	1
F12_S09	10	high	1.000000	35389440	0	1
F12_S10	10	high	1.000000	35389440	0	1
F12_S11	10	high	1.000000	35389440	0	1
F12_S12	10	high	1.000000	35389440	0	1
F12_S01	30	low	0.322222	11476992	87398	28
F12_S02	30	low	0.322222	11476992	87398	28
F12_S03	30	low	0.322222	11476992	87398	28
F12_S04	30	low	0.322222	11476992	87398	28
F12_S05	30	low	0.322222	11476992	87398	28
F12_S06	30	low	0.322222	11476992	87398	28
F12_S07	30	low	0.322222	11476992	87398	28
F12_S08	30	low	0.322222	11476992	87398	28

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S09	30	low	0.322222	11476992	87398	28
F12_S10	30	low	0.322222	11476992	87398	28
F12_S11	30	low	0.322222	11476992	87398	28
F12_S12	30	low	0.322222	11476992	87398	28
F12_S01	30	mid	0.500000	17694720	43691	2
F12_S02	30	mid	0.500000	17694720	43691	2
F12_S03	30	mid	0.500000	17694720	43691	2
F12_S04	30	mid	0.500000	17694720	43691	2
F12_S05	30	mid	0.500000	17694720	43691	2
F12_S06	30	mid	0.500000	17694720	43691	2
F12_S07	30	mid	0.500000	17694720	43691	2
F12_S08	30	mid	0.500000	17694720	43691	2
F12_S09	30	mid	0.500000	17694720	43691	2
F12_S10	30	mid	0.500000	17694720	43691	2
F12_S11	30	mid	0.500000	17694720	43691	2
F12_S12	30	mid	0.500000	17694720	43691	2
F12_S01	30	high	1.000000	35389440	0	1
F12_S02	30	high	1.000000	35389440	0	1
F12_S03	30	high	1.000000	35389440	0	1
F12_S04	30	high	1.000000	35389440	0	1
F12_S05	30	high	1.000000	35389440	0	1
F12_S06	30	high	1.000000	35389440	0	1
F12_S07	30	high	1.000000	35389440	0	1
F12_S08	30	high	1.000000	35389440	0	1
F12_S09	30	high	1.000000	35389440	0	1
F12_S10	30	high	1.000000	35389440	0	1
F12_S11	30	high	1.000000	35389440	0	1
F12_S12	30	high	1.000000	35389440	0	1
F12_S01	100	low	0.333333	11821056	87436	—
F12_S02	100	low	0.333333	11821056	87436	—
F12_S03	100	low	0.333333	11821056	87436	—
F12_S04	100	low	0.333333	11821056	87436	—
F12_S05	100	low	0.333333	11821056	87436	—
F12_S06	100	low	0.333333	11821056	87436	—
F12_S07	100	low	0.333333	11821056	87436	—
F12_S08	100	low	0.333333	11821056	87436	—
F12_S09	100	low	0.333333	11821056	87436	—
F12_S10	100	low	0.333333	11821056	87436	—
F12_S11	100	low	0.333333	11821056	87436	—
F12_S12	100	low	0.333333	11821056	87436	—
F12_S01	100	mid	0.500000	17694720	43691	2
F12_S02	100	mid	0.500000	17694720	43691	2
F12_S03	100	mid	0.500000	17694720	43691	2
F12_S04	100	mid	0.500000	17694720	43691	2
F12_S05	100	mid	0.500000	17694720	43691	2
F12_S06	100	mid	0.500000	17694720	43691	2
F12_S07	100	mid	0.500000	17694720	43691	2
F12_S08	100	mid	0.500000	17694720	43691	2
F12_S09	100	mid	0.500000	17694720	43691	2
F12_S10	100	mid	0.500000	17694720	43691	2
F12_S11	100	mid	0.500000	17694720	43691	2
F12_S12	100	mid	0.500000	17694720	43691	2
F12_S01	100	high	1.000000	35389440	0	1
F12_S02	100	high	1.000000	35389440	0	1
F12_S03	100	high	1.000000	35389440	0	1
F12_S04	100	high	1.000000	35389440	0	1
F12_S05	100	high	1.000000	35389440	0	1
F12_S06	100	high	1.000000	35389440	0	1
F12_S07	100	high	1.000000	35389440	0	1
F12_S08	100	high	1.000000	35389440	0	1
F12_S09	100	high	1.000000	35389440	0	1
F12_S10	100	high	1.000000	35389440	0	1

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S11	100	high	1.000000	35389440	0	1
F12_S12	100	high	1.000000	35389440	0	1
F12_S01	168	low	0.244444	9209088	87382	4
F12_S02	168	low	0.244444	9209088	87382	4
F12_S03	168	low	0.244444	9209088	87382	4
F12_S04	168	low	0.244444	9209088	87382	4
F12_S05	168	low	0.244444	9209088	87382	4
F12_S06	168	low	0.244444	9209088	87382	4
F12_S07	168	low	0.244444	9209088	87382	4
F12_S08	168	low	0.244444	9209088	87382	4
F12_S09	168	low	0.244444	9209088	87382	4
F12_S10	168	low	0.244444	9209088	87382	4
F12_S11	168	low	0.244444	9209088	87382	4
F12_S12	168	low	0.244444	9209088	87382	4
F12_S01	168	mid	0.500000	17729280	43691	2
F12_S02	168	mid	0.500000	17729280	43691	2
F12_S03	168	mid	0.500000	17729280	43691	2
F12_S04	168	mid	0.500000	17729280	43691	2
F12_S05	168	mid	0.500000	17729280	43691	2
F12_S06	168	mid	0.500000	17729280	43691	2
F12_S07	168	mid	0.500000	17729280	43691	2
F12_S08	168	mid	0.500000	17729280	43691	2
F12_S09	168	mid	0.500000	17729280	43691	2
F12_S10	168	mid	0.500000	17729280	43691	2
F12_S11	168	mid	0.500000	17729280	43691	2
F12_S12	168	mid	0.500000	17729280	43691	2
F12_S01	168	high	1.000000	35389440	0	1
F12_S02	168	high	1.000000	35389440	0	1
F12_S03	168	high	1.000000	35389440	0	1
F12_S04	168	high	1.000000	35389440	0	1
F12_S05	168	high	1.000000	35389440	0	1
F12_S06	168	high	1.000000	35389440	0	1
F12_S07	168	high	1.000000	35389440	0	1
F12_S08	168	high	1.000000	35389440	0	1
F12_S09	168	high	1.000000	35389440	0	1
F12_S10	168	high	1.000000	35389440	0	1
F12_S11	168	high	1.000000	35389440	0	1
F12_S12	168	high	1.000000	35389440	0	1
F12_S01	300	low	0.244444	9209088	87382	4
F12_S02	300	low	0.244444	9209088	87382	4
F12_S03	300	low	0.244444	9209088	87382	4
F12_S04	300	low	0.244444	9209088	87382	4
F12_S05	300	low	0.244444	9209088	87382	4
F12_S06	300	low	0.244444	9209088	87382	4
F12_S07	300	low	0.244444	9209088	87382	4
F12_S08	300	low	0.244444	9209088	87382	4
F12_S09	300	low	0.244444	9209088	87382	4
F12_S10	300	low	0.244444	9209088	87382	4
F12_S11	300	low	0.244444	9209088	87382	4
F12_S12	300	low	0.244444	9209088	87382	4
F12_S01	300	mid	0.500000	17729280	43691	2
F12_S02	300	mid	0.500000	17729280	43691	2
F12_S03	300	mid	0.500000	17729280	43691	2
F12_S04	300	mid	0.500000	17729280	43691	2
F12_S05	300	mid	0.500000	17729280	43691	2
F12_S06	300	mid	0.500000	17729280	43691	2
F12_S07	300	mid	0.500000	17729280	43691	2
F12_S08	300	mid	0.500000	17729280	43691	2
F12_S09	300	mid	0.500000	17729280	43691	2
F12_S10	300	mid	0.500000	17729280	43691	2
F12_S11	300	mid	0.500000	17729280	43691	2
F12_S12	300	mid	0.500000	17729280	43691	2

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S01	300	high	1.000000	35389440	0	1
F12_S02	300	high	1.000000	35389440	0	1
F12_S03	300	high	1.000000	35389440	0	1
F12_S04	300	high	1.000000	35389440	0	1
F12_S05	300	high	1.000000	35389440	0	1
F12_S06	300	high	1.000000	35389440	0	1
F12_S07	300	high	1.000000	35389440	0	1
F12_S08	300	high	1.000000	35389440	0	1
F12_S09	300	high	1.000000	35389440	0	1
F12_S10	300	high	1.000000	35389440	0	1
F12_S11	300	high	1.000000	35389440	0	1
F12_S12	300	high	1.000000	35389440	0	1
F12_S01	1000	low	0.288889	10582272	87896	7
F12_S02	1000	low	0.288889	10582272	87896	7
F12_S03	1000	low	0.288889	10582272	87896	7
F12_S04	1000	low	0.288889	10582272	87896	7
F12_S05	1000	low	0.288889	10582272	87896	7
F12_S06	1000	low	0.288889	10582272	87896	7
F12_S07	1000	low	0.288889	10582272	87896	7
F12_S08	1000	low	0.288889	10582272	87896	7
F12_S09	1000	low	0.288889	10582272	87896	7
F12_S10	1000	low	0.288889	10582272	87896	7
F12_S11	1000	low	0.288889	10582272	87896	7
F12_S12	1000	low	0.288889	10582272	87896	7
F12_S01	1000	mid	0.500000	17729280	43691	2
F12_S02	1000	mid	0.500000	17729280	43691	2
F12_S03	1000	mid	0.500000	17729280	43691	2
F12_S04	1000	mid	0.500000	17729280	43691	2
F12_S05	1000	mid	0.500000	17729280	43691	2
F12_S06	1000	mid	0.500000	17729280	43691	2
F12_S07	1000	mid	0.500000	17729280	43691	2
F12_S08	1000	mid	0.500000	17729280	43691	2
F12_S09	1000	mid	0.500000	17729280	43691	2
F12_S10	1000	mid	0.500000	17729280	43691	2
F12_S11	1000	mid	0.500000	17729280	43691	2
F12_S12	1000	mid	0.500000	17729280	43691	2
F12_S01	1000	high	1.000000	35389440	0	1
F12_S02	1000	high	1.000000	35389440	0	1
F12_S03	1000	high	1.000000	35389440	0	1
F12_S04	1000	high	1.000000	35389440	0	1
F12_S05	1000	high	1.000000	35389440	0	1
F12_S06	1000	high	1.000000	35389440	0	1
F12_S07	1000	high	1.000000	35389440	0	1
F12_S08	1000	high	1.000000	35389440	0	1
F12_S09	1000	high	1.000000	35389440	0	1
F12_S10	1000	high	1.000000	35389440	0	1
F12_S11	1000	high	1.000000	35389440	0	1
F12_S12	1000	high	1.000000	35389440	0	1
F12_S01	3000	low	0.244444	9491712	87382	4
F12_S02	3000	low	0.244444	9491712	87382	4
F12_S03	3000	low	0.244444	9491712	87382	4
F12_S04	3000	low	0.244444	9491712	87382	4
F12_S05	3000	low	0.244444	9491712	87382	4
F12_S06	3000	low	0.244444	9491712	87382	4
F12_S07	3000	low	0.244444	9491712	87382	4
F12_S08	3000	low	0.244444	9491712	87382	4
F12_S09	3000	low	0.244444	9491712	87382	4
F12_S10	3000	low	0.244444	9491712	87382	4
F12_S11	3000	low	0.244444	9491712	87382	4
F12_S12	3000	low	0.244444	9491712	87382	4
F12_S01	3000	mid	0.500000	18247680	43691	2
F12_S02	3000	mid	0.500000	18247680	43691	2

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S03	3000	mid	0.500000	18247680	43691	2
F12_S04	3000	mid	0.500000	18247680	43691	2
F12_S05	3000	mid	0.500000	18247680	43691	2
F12_S06	3000	mid	0.500000	18247680	43691	2
F12_S07	3000	mid	0.500000	18247680	43691	2
F12_S08	3000	mid	0.500000	18247680	43691	2
F12_S09	3000	mid	0.500000	18247680	43691	2
F12_S10	3000	mid	0.500000	18247680	43691	2
F12_S11	3000	mid	0.500000	18247680	43691	2
F12_S12	3000	mid	0.500000	18247680	43691	2
F12_S01	3000	high	1.000000	35389440	0	1
F12_S02	3000	high	1.000000	35389440	0	1
F12_S03	3000	high	1.000000	35389440	0	1
F12_S04	3000	high	1.000000	35389440	0	1
F12_S05	3000	high	1.000000	35389440	0	1
F12_S06	3000	high	1.000000	35389440	0	1
F12_S07	3000	high	1.000000	35389440	0	1
F12_S08	3000	high	1.000000	35389440	0	1
F12_S09	3000	high	1.000000	35389440	0	1
F12_S10	3000	high	1.000000	35389440	0	1
F12_S11	3000	high	1.000000	35389440	0	1
F12_S12	3000	high	1.000000	35389440	0	1
F12_S01	10000	low	0.244444	9756672	87382	4
F12_S02	10000	low	0.244444	9756672	87382	4
F12_S03	10000	low	0.244444	9756672	87382	4
F12_S04	10000	low	0.244444	9756672	87382	4
F12_S05	10000	low	0.244444	9756672	87382	4
F12_S06	10000	low	0.244444	9756672	87382	4
F12_S07	10000	low	0.244444	9756672	87382	4
F12_S08	10000	low	0.244444	9756672	87382	4
F12_S09	10000	low	0.244444	9756672	87382	4
F12_S10	10000	low	0.244444	9756672	87382	4
F12_S11	10000	low	0.244444	9756672	87382	4
F12_S12	10000	low	0.244444	9756672	87382	4
F12_S01	10000	mid	0.500000	18247680	43691	2
F12_S02	10000	mid	0.500000	18247680	43691	2
F12_S03	10000	mid	0.500000	18247680	43691	2
F12_S04	10000	mid	0.500000	18247680	43691	2
F12_S05	10000	mid	0.500000	18247680	43691	2
F12_S06	10000	mid	0.500000	18247680	43691	2
F12_S07	10000	mid	0.500000	18247680	43691	2
F12_S08	10000	mid	0.500000	18247680	43691	2
F12_S09	10000	mid	0.500000	18247680	43691	2
F12_S10	10000	mid	0.500000	18247680	43691	2
F12_S11	10000	mid	0.500000	18247680	43691	2
F12_S12	10000	mid	0.500000	18247680	43691	2
F12_S01	10000	high	1.000000	35389440	0	1
F12_S02	10000	high	1.000000	35389440	0	1
F12_S03	10000	high	1.000000	35389440	0	1
F12_S04	10000	high	1.000000	35389440	0	1
F12_S05	10000	high	1.000000	35389440	0	1
F12_S06	10000	high	1.000000	35389440	0	1
F12_S07	10000	high	1.000000	35389440	0	1
F12_S08	10000	high	1.000000	35389440	0	1
F12_S09	10000	high	1.000000	35389440	0	1
F12_S10	10000	high	1.000000	35389440	0	1
F12_S11	10000	high	1.000000	35389440	0	1
F12_S12	10000	high	1.000000	35389440	0	1
F12_S01	30000	low	0.288889	11489280	99672	7
F12_S02	30000	low	0.288889	11489280	99672	7
F12_S03	30000	low	0.288889	11489280	99672	7
F12_S04	30000	low	0.288889	11489280	99672	7

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S05	30000	low	0.288889	11489280	99672	7
F12_S06	30000	low	0.288889	11489280	99672	7
F12_S07	30000	low	0.288889	11489280	99672	7
F12_S08	30000	low	0.288889	11489280	99672	7
F12_S09	30000	low	0.288889	11489280	99672	7
F12_S10	30000	low	0.288889	11489280	99672	7
F12_S11	30000	low	0.288889	11489280	99672	7
F12_S12	30000	low	0.288889	11489280	99672	7
F12_S01	30000	mid	0.500000	18800640	43691	2
F12_S02	30000	mid	0.500000	18800640	43691	2
F12_S03	30000	mid	0.500000	18800640	43691	2
F12_S04	30000	mid	0.500000	18800640	43691	2
F12_S05	30000	mid	0.500000	18800640	43691	2
F12_S06	30000	mid	0.500000	18800640	43691	2
F12_S07	30000	mid	0.500000	18800640	43691	2
F12_S08	30000	mid	0.500000	18800640	43691	2
F12_S09	30000	mid	0.500000	18800640	43691	2
F12_S10	30000	mid	0.500000	18800640	43691	2
F12_S11	30000	mid	0.500000	18800640	43691	2
F12_S12	30000	mid	0.500000	18800640	43691	2
F12_S01	30000	high	1.000000	35389440	0	1
F12_S02	30000	high	1.000000	35389440	0	1
F12_S03	30000	high	1.000000	35389440	0	1
F12_S04	30000	high	1.000000	35389440	0	1
F12_S05	30000	high	1.000000	35389440	0	1
F12_S06	30000	high	1.000000	35389440	0	1
F12_S07	30000	high	1.000000	35389440	0	1
F12_S08	30000	high	1.000000	35389440	0	1
F12_S09	30000	high	1.000000	35389440	0	1
F12_S10	30000	high	1.000000	35389440	0	1
F12_S11	30000	high	1.000000	35389440	0	1
F12_S12	30000	high	1.000000	35389440	0	1
F12_S01	100000	low	0.288889	11796480	142052	—
F12_S02	100000	low	0.288889	11796480	142052	—
F12_S03	100000	low	0.288889	11796480	142052	—
F12_S04	100000	low	0.288889	11796480	142052	—
F12_S05	100000	low	0.288889	11796480	142052	—
F12_S06	100000	low	0.288889	11796480	142052	—
F12_S07	100000	low	0.288889	11796480	142052	—
F12_S08	100000	low	0.288889	11796480	142052	—
F12_S09	100000	low	0.288889	11796480	142052	—
F12_S10	100000	low	0.288889	11796480	142052	—
F12_S11	100000	low	0.288889	11796480	142052	—
F12_S12	100000	low	0.288889	11796480	142052	—
F12_S01	100000	mid	0.622222	22855680	95575	8
F12_S02	100000	mid	0.622222	22855680	95575	8
F12_S03	100000	mid	0.622222	22855680	95575	8
F12_S04	100000	mid	0.622222	22855680	95575	8
F12_S05	100000	mid	0.622222	22855680	95575	8
F12_S06	100000	mid	0.622222	22855680	95575	8
F12_S07	100000	mid	0.622222	22855680	95575	8
F12_S08	100000	mid	0.622222	22855680	95575	8
F12_S09	100000	mid	0.622222	22855680	95575	8
F12_S10	100000	mid	0.622222	22855680	95575	8
F12_S11	100000	mid	0.622222	22855680	95575	8
F12_S12	100000	mid	0.622222	22855680	95575	8
F12_S01	100000	high	1.000000	35389440	0	1
F12_S02	100000	high	1.000000	35389440	0	1
F12_S03	100000	high	1.000000	35389440	0	1
F12_S04	100000	high	1.000000	35389440	0	1
F12_S05	100000	high	1.000000	35389440	0	1
F12_S06	100000	high	1.000000	35389440	0	1

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S07	100000	high	1.000000	35389440	0	1
F12_S08	100000	high	1.000000	35389440	0	1
F12_S09	100000	high	1.000000	35389440	0	1
F12_S10	100000	high	1.000000	35389440	0	1
F12_S11	100000	high	1.000000	35389440	0	1
F12_S12	100000	high	1.000000	35389440	0	1

V. Complete Desynchronization Sensitivity

Probe phases are independently assigned from fixed seeds. Probes exchange neither action nor queue state, and no joint admission is introduced.

TABLE IV: Complete desynchronization software-sensitivity results (240 retained rows).

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090901	low	desynchronized_independent	0.333333	0.000000	131029	595/193/691
2026090901	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090901	low	central_coordinated	0.000000	–	16341	0/0/0
2026090901	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090901	mid	desynchronized_independent	0.666667	0.505556	130985	595/193/691
2026090901	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090901	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090901	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090901	high	desynchronized_independent	1.000000	1.000000	121111	595/193/691
2026090901	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090901	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090901	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090902	low	desynchronized_independent	0.333333	0.000000	131029	403/632/306
2026090902	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090902	low	central_coordinated	0.000000	–	16341	0/0/0
2026090902	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090902	mid	desynchronized_independent	0.666667	0.500000	130985	403/632/306
2026090902	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090902	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090902	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090902	high	desynchronized_independent	1.000000	1.000000	130876	403/632/306
2026090902	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090902	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090902	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090903	low	desynchronized_independent	0.333333	0.000000	131029	489/690/996
2026090903	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090903	low	central_coordinated	0.000000	–	16341	0/0/0
2026090903	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090903	mid	desynchronized_independent	0.666667	0.500000	130985	489/690/996
2026090903	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090903	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090903	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090903	high	desynchronized_independent	1.000000	1.000000	1180	489/690/996
2026090903	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090903	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090903	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090904	low	desynchronized_independent	0.333333	0.000000	131028	945/680/799
2026090904	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090904	low	central_coordinated	0.000000	–	16341	0/0/0
2026090904	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090904	mid	desynchronized_independent	0.666667	0.500000	130985	945/680/799
2026090904	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090904	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090904	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090904	high	desynchronized_independent	1.000000	1.000000	21234	945/680/799
2026090904	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090904	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090904	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090905	low	desynchronized_independent	0.333333	0.000000	131028	154/172/828
2026090905	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090905	low	central_coordinated	0.000000	–	16341	0/0/0
2026090905	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090905	mid	desynchronized_independent	0.666667	0.505556	130985	154/172/828
2026090905	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090905	mid	central_coordinated	0.333333	1.000000	130985	0/0/0

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090905	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090905	high	desynchronized_independent	1.000000	1.000000	130876	154/172/828
2026090905	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090905	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090905	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090906	low	desynchronized_independent	0.333333	0.000000	131029	60/626/317
2026090906	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090906	low	central_coordinated	0.000000	–	16341	0/0/0
2026090906	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090906	mid	desynchronized_independent	0.666667	0.505556	130985	60/626/317
2026090906	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090906	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090906	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090906	high	desynchronized_independent	1.000000	1.000000	130876	60/626/317
2026090906	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090906	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090906	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090907	low	desynchronized_independent	0.333333	0.000000	131029	753/955/576
2026090907	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090907	low	central_coordinated	0.000000	–	16341	0/0/0
2026090907	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090907	mid	desynchronized_independent	0.666667	0.500000	130985	753/955/576
2026090907	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090907	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090907	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090907	high	desynchronized_independent	1.000000	1.000000	17302	753/955/576
2026090907	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090907	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090907	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090908	low	desynchronized_independent	0.333333	0.000000	131029	884/483/389
2026090908	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090908	low	central_coordinated	0.000000	–	16341	0/0/0
2026090908	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090908	mid	desynchronized_independent	0.666667	0.500000	130985	884/483/389
2026090908	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090908	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090908	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090908	high	desynchronized_independent	1.000000	1.000000	71828	884/483/389
2026090908	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090908	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090908	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090909	low	desynchronized_independent	0.333333	0.000000	131029	530/348/834
2026090909	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090909	low	central_coordinated	0.000000	–	16341	0/0/0
2026090909	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090909	mid	desynchronized_independent	0.666667	0.500000	130985	530/348/834
2026090909	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090909	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090909	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090909	high	desynchronized_independent	1.000000	1.000000	64881	530/348/834
2026090909	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090909	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090909	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090910	low	desynchronized_independent	0.333333	0.000000	131028	611/131/96
2026090910	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090910	low	central_coordinated	0.000000	–	16341	0/0/0
2026090910	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090910	mid	desynchronized_independent	0.666667	0.505556	130985	611/131/96
2026090910	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090910	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090910	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090910	high	desynchronized_independent	1.000000	1.000000	130876	611/131/96

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090910	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090910	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090910	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090911	low	desynchronized_independent	0.333333	0.000000	131029	286/697/613
2026090911	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090911	low	central_coordinated	0.000000	–	16341	0/0/0
2026090911	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090911	mid	desynchronized_independent	0.666667	0.505556	130985	286/697/613
2026090911	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090911	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090911	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090911	high	desynchronized_independent	1.000000	1.000000	118752	286/697/613
2026090911	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090911	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090911	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090912	low	desynchronized_independent	0.333333	0.000000	131028	72/898/820
2026090912	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090912	low	central_coordinated	0.000000	–	16341	0/0/0
2026090912	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090912	mid	desynchronized_independent	0.666667	0.505556	130985	72/898/820
2026090912	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090912	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090912	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090912	high	desynchronized_independent	1.000000	1.000000	102368	72/898/820
2026090912	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090912	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090912	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090913	low	desynchronized_independent	0.333333	0.000000	131028	31/997/848
2026090913	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090913	low	central_coordinated	0.000000	–	16341	0/0/0
2026090913	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090913	mid	desynchronized_independent	0.666667	0.505556	130985	31/997/848
2026090913	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090913	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090913	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090913	high	desynchronized_independent	1.000000	1.000000	118490	31/997/848
2026090913	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090913	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090913	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090914	low	desynchronized_independent	0.333333	0.000000	131029	379/396/678
2026090914	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090914	low	central_coordinated	0.000000	–	16341	0/0/0
2026090914	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090914	mid	desynchronized_independent	0.666667	0.500000	130985	379/396/678
2026090914	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090914	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090914	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090914	high	desynchronized_independent	1.000000	1.000000	126223	379/396/678
2026090914	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090914	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090914	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090915	low	desynchronized_independent	0.333333	0.000000	131028	212/604/213
2026090915	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090915	low	central_coordinated	0.000000	–	16341	0/0/0
2026090915	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090915	mid	desynchronized_independent	0.666667	0.505556	130985	212/604/213
2026090915	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090915	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090915	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090915	high	desynchronized_independent	1.000000	1.000000	130876	212/604/213
2026090915	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090915	high	central_coordinated	1.000000	1.000000	130941	0/0/0

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090915	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090916	low	desynchronized_independent	0.333333	0.000000	131029	586/124/90
2026090916	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090916	low	central_coordinated	0.000000	–	16341	0/0/0
2026090916	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090916	mid	desynchronized_independent	0.666667	0.505556	130985	586/124/90
2026090916	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090916	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090916	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090916	high	desynchronized_independent	1.000000	1.000000	130876	586/124/90
2026090916	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090916	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090916	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090917	low	desynchronized_independent	0.333333	0.000000	131029	13/5/654
2026090917	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090917	low	central_coordinated	0.000000	–	16341	0/0/0
2026090917	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090917	mid	desynchronized_independent	0.666667	0.500000	130985	13/5/654
2026090917	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090917	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090917	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090917	high	desynchronized_independent	1.000000	1.000000	130876	13/5/654
2026090917	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090917	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090917	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090918	low	desynchronized_independent	0.333333	0.000000	131029	303/966/432
2026090918	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090918	low	central_coordinated	0.000000	–	16341	0/0/0
2026090918	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090918	mid	desynchronized_independent	0.666667	0.500000	130985	303/966/432
2026090918	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090918	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090918	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090918	high	desynchronized_independent	1.000000	1.000000	91882	303/966/432
2026090918	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090918	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090918	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090919	low	desynchronized_independent	0.333333	0.000000	131029	94/427/937
2026090919	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090919	low	central_coordinated	0.000000	–	16341	0/0/0
2026090919	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090919	mid	desynchronized_independent	0.666667	0.500000	130985	94/427/937
2026090919	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090919	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090919	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090919	high	desynchronized_independent	1.000000	1.000000	93848	94/427/937
2026090919	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090919	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090919	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090920	low	desynchronized_independent	0.333333	0.000000	131029	699/148/926
2026090920	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090920	low	central_coordinated	0.000000	–	16341	0/0/0
2026090920	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090920	mid	desynchronized_independent	0.666667	0.505556	130985	699/148/926
2026090920	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090920	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090920	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090920	high	desynchronized_independent	1.000000	1.000000	72483	699/148/926
2026090920	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090920	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090920	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0

VI. Complete Nominal-Deadline Reconstruction

Post-release latency is result return minus actual event creation. Nominal latency is result return minus scheduled release. The 1-s quantity in the main paper is a post-release service SLO, not a complete schedule-to-result deadline.

TABLE V: Complete accepted-hardware timing reconstruction (648 retained cells; latency in ms).

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S01	fixed_local	low	P1	0	–	–	1862.2	1920.9	1924.1
F12_S01	fixed_local	low	P2	0	–	–	760.3	831.8	834.1
F12_S01	fixed_local	low	P3	0	–	–	86.3	155.7	162.0
F12_S01	fixed_local	mid	P1	0	–	–	1828.4	1979.1	1980.6
F12_S01	fixed_local	mid	P2	0	–	–	636.8	775.7	780.4
F12_S01	fixed_local	mid	P3	0	–	–	87.5	165.1	172.5
F12_S01	fixed_local	high	P1	0	–	–	1827.5	1863.0	1913.4
F12_S01	fixed_local	high	P2	0	–	–	637.0	671.0	673.3
F12_S01	fixed_local	high	P3	0	–	–	84.4	156.5	163.5
F12_S01	fixed_summary	low	P1	0	–	–	1952.6	2002.0	2006.5
F12_S01	fixed_summary	low	P2	0	–	–	712.1	750.1	753.4
F12_S01	fixed_summary	low	P3	0	–	–	84.0	156.4	163.8
F12_S01	fixed_summary	mid	P1	0	–	–	1922.6	1997.1	2001.5
F12_S01	fixed_summary	mid	P2	0	–	–	716.0	804.2	806.5
F12_S01	fixed_summary	mid	P3	0	–	–	85.2	153.6	160.0
F12_S01	fixed_summary	high	P1	0	–	–	1925.6	1978.0	1981.4
F12_S01	fixed_summary	high	P2	0	–	–	745.5	802.9	806.8
F12_S01	fixed_summary	high	P3	0	–	–	88.3	159.6	166.3
F12_S01	fixed_full	low	P1	90	0.0000	0.0000	1887.7	1981.0	1983.3
F12_S01	fixed_full	low	P2	90	0.0000	0.0000	736.5	827.9	830.4
F12_S01	fixed_full	low	P3	90	0.0000	0.0000	79.3	139.5	145.3
F12_S01	fixed_full	mid	P1	90	0.0000	0.0000	1859.1	2483.1	2486.7
F12_S01	fixed_full	mid	P2	90	0.0000	0.0000	686.7	1341.8	1346.6
F12_S01	fixed_full	mid	P3	90	0.0000	0.0000	79.2	141.9	148.5
F12_S01	fixed_full	high	P1	90	0.9667	0.0000	2024.3	2101.8	2105.2
F12_S01	fixed_full	high	P2	90	0.9222	0.0000	830.5	948.8	952.3
F12_S01	fixed_full	high	P3	90	1.0000	1.0000	78.6	141.2	145.6
F12_S01	independent_dpp	low	P1	22	0.0000	0.0000	1851.5	2183.8	2301.2
F12_S01	independent_dpp	low	P2	22	0.0000	0.0000	661.8	792.4	1053.9
F12_S01	independent_dpp	low	P3	22	0.0000	0.0000	82.3	152.1	156.3
F12_S01	independent_dpp	mid	P1	45	0.0000	0.0000	1998.0	2108.8	2111.0
F12_S01	independent_dpp	mid	P2	45	0.0000	0.0000	754.9	822.0	824.8
F12_S01	independent_dpp	mid	P3	45	0.0000	0.0000	81.4	146.9	152.4
F12_S01	independent_dpp	high	P1	90	0.9444	0.0000	1967.9	2006.6	2009.6
F12_S01	independent_dpp	high	P2	90	0.9556	0.0000	797.6	836.1	841.8
F12_S01	independent_dpp	high	P3	90	1.0000	1.0000	78.4	136.1	140.8
F12_S01	central_dpp	low	P1	0	–	–	2049.7	2151.4	2153.5
F12_S01	central_dpp	low	P2	0	–	–	858.4	942.6	947.1
F12_S01	central_dpp	low	P3	0	–	–	83.2	148.0	154.0
F12_S01	central_dpp	mid	P1	30	0.8000	0.0000	1910.4	1992.9	1995.4
F12_S01	central_dpp	mid	P2	30	0.8000	0.0000	732.4	809.6	813.0
F12_S01	central_dpp	mid	P3	30	1.0000	1.0000	83.1	152.5	159.3
F12_S01	central_dpp	high	P1	90	1.0000	0.0000	2070.5	2105.4	2109.1
F12_S01	central_dpp	high	P2	90	0.9889	0.0000	907.5	949.5	953.2
F12_S01	central_dpp	high	P3	90	1.0000	1.0000	78.5	144.9	150.8
F12_S01	shielded_dpp	low	P1	0	–	–	1771.7	1832.4	1835.5
F12_S01	shielded_dpp	low	P2	0	–	–	678.1	749.6	786.5
F12_S01	shielded_dpp	low	P3	0	–	–	87.8	156.7	162.3
F12_S01	shielded_dpp	mid	P1	30	0.9000	0.0000	1934.5	2160.6	2163.7
F12_S01	shielded_dpp	mid	P2	30	1.0000	0.0000	856.4	964.9	967.9
F12_S01	shielded_dpp	mid	P3	30	1.0000	1.0000	86.3	149.8	155.9
F12_S01	shielded_dpp	high	P1	90	0.9333	0.0000	1979.2	2065.8	2070.3
F12_S01	shielded_dpp	high	P2	90	0.9444	0.0000	876.4	913.1	974.2
F12_S01	shielded_dpp	high	P3	90	1.0000	1.0000	79.7	136.0	140.8
F12_S02	fixed_local	low	P1	0	–	–	1886.4	2021.6	2024.1
F12_S02	fixed_local	low	P2	0	–	–	594.0	740.2	743.2
F12_S02	fixed_local	low	P3	0	–	–	88.2	164.6	171.5
F12_S02	fixed_local	mid	P1	0	–	–	2003.6	2046.7	2051.4
F12_S02	fixed_local	mid	P2	0	–	–	717.1	761.0	764.1
F12_S02	fixed_local	mid	P3	0	–	–	92.2	161.7	168.4
F12_S02	fixed_local	high	P1	0	–	–	1896.5	1938.2	1941.3
F12_S02	fixed_local	high	P2	0	–	–	737.1	775.7	778.9
F12_S02	fixed_local	high	P3	0	–	–	91.3	160.5	167.6
F12_S02	fixed_summary	low	P1	0	–	–	1725.1	1764.9	1767.8
F12_S02	fixed_summary	low	P2	0	–	–	550.6	585.7	589.6
F12_S02	fixed_summary	low	P3	0	–	–	90.6	160.8	167.3
F12_S02	fixed_summary	mid	P1	0	–	–	1915.3	2065.5	2072.1
F12_S02	fixed_summary	mid	P2	0	–	–	677.3	1103.5	1106.6
F12_S02	fixed_summary	mid	P3	0	–	–	85.8	156.6	162.7
F12_S02	fixed_summary	high	P1	0	–	–	1878.3	1949.3	1959.8
F12_S02	fixed_summary	high	P2	0	–	–	713.1	787.0	798.8
F12_S02	fixed_summary	high	P3	0	–	–	89.0	158.6	165.8
F12_S02	fixed_full	low	P1	90	0.0000	0.0000	1929.0	2110.4	2114.6
F12_S02	fixed_full	low	P2	90	0.0000	0.0000	763.0	876.5	879.5

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S02	fixed_full	low	P3	90	0.0000	0.0000	78.2	135.9	142.5
F12_S02	fixed_full	mid	P1	90	0.0000	0.0000	2251.6	2305.8	2309.9
F12_S02	fixed_full	mid	P2	90	0.0000	0.0000	616.5	733.8	750.5
F12_S02	fixed_full	mid	P3	90	0.0000	0.0000	76.4	136.6	142.5
F12_S02	fixed_full	high	P1	90	1.0000	0.0000	1969.5	1996.6	2001.5
F12_S02	fixed_full	high	P2	90	0.9889	0.0000	827.2	865.1	867.9
F12_S02	fixed_full	high	P3	90	1.0000	1.0000	80.0	145.5	151.7
F12_S02	independent_dpp	low	P1	22	0.0000	0.0000	1906.4	2060.6	2064.0
F12_S02	independent_dpp	low	P2	22	0.0000	0.0000	748.0	899.5	904.3
F12_S02	independent_dpp	low	P3	22	0.0000	0.0000	78.9	142.0	147.5
F12_S02	independent_dpp	mid	P1	45	0.0000	0.0000	1980.3	2013.9	2015.6
F12_S02	independent_dpp	mid	P2	45	0.0000	0.0000	703.8	732.1	734.0
F12_S02	independent_dpp	mid	P3	45	0.0000	0.0000	79.5	147.8	152.5
F12_S02	independent_dpp	high	P1	90	0.9111	0.0000	1982.0	2052.8	2056.0
F12_S02	independent_dpp	high	P2	90	0.9333	0.0000	787.2	831.0	832.9
F12_S02	independent_dpp	high	P3	90	1.0000	1.0000	78.0	137.5	141.7
F12_S02	central_dpp	low	P1	0	—	—	2017.6	2154.2	2157.5
F12_S02	central_dpp	low	P2	0	—	—	738.5	861.7	865.2
F12_S02	central_dpp	low	P3	0	—	—	87.9	158.3	164.1
F12_S02	central_dpp	mid	P1	30	0.7333	0.0000	1936.7	2071.6	2074.8
F12_S02	central_dpp	mid	P2	30	0.9333	0.0000	881.1	913.6	916.2
F12_S02	central_dpp	mid	P3	30	1.0000	1.0000	86.2	150.4	156.4
F12_S02	central_dpp	high	P1	90	0.9222	0.0000	1942.0	1995.4	2000.0
F12_S02	central_dpp	high	P2	90	0.9444	0.0000	816.7	862.3	865.6
F12_S02	central_dpp	high	P3	90	1.0000	1.0000	81.3	145.8	151.8
F12_S02	shielded_dpp	low	P1	0	—	—	1931.4	1995.7	1999.1
F12_S02	shielded_dpp	low	P2	0	—	—	842.4	903.9	908.0
F12_S02	shielded_dpp	low	P3	0	—	—	86.6	160.6	168.3
F12_S02	shielded_dpp	mid	P1	30	0.9000	0.0000	2120.6	2150.6	2151.7
F12_S02	shielded_dpp	mid	P2	30	0.9000	0.0000	793.9	826.4	828.9
F12_S02	shielded_dpp	mid	P3	30	1.0000	1.0000	80.3	148.0	155.9
F12_S02	shielded_dpp	high	P1	90	0.9444	0.0000	1954.9	2007.4	2039.4
F12_S02	shielded_dpp	high	P2	90	0.9333	0.0000	781.0	858.6	873.6
F12_S02	shielded_dpp	high	P3	90	1.0000	1.0000	81.5	140.4	145.8
F12_S03	fixed_local	low	P1	0	—	—	1892.3	1942.8	1950.8
F12_S03	fixed_local	low	P2	0	—	—	676.8	732.8	744.0
F12_S03	fixed_local	low	P3	0	—	—	90.0	162.6	169.3
F12_S03	fixed_local	mid	P1	0	—	—	1823.3	1868.5	1871.1
F12_S03	fixed_local	mid	P2	0	—	—	736.6	837.7	840.2
F12_S03	fixed_local	mid	P3	0	—	—	89.2	160.3	166.4
F12_S03	fixed_local	high	P1	0	—	—	1939.5	2052.7	2061.3
F12_S03	fixed_local	high	P2	0	—	—	724.4	838.4	852.7
F12_S03	fixed_local	high	P3	0	—	—	90.8	167.3	174.3
F12_S03	fixed_summary	low	P1	0	—	—	1926.4	1967.6	1971.1
F12_S03	fixed_summary	low	P2	0	—	—	710.5	757.2	762.6
F12_S03	fixed_summary	low	P3	0	—	—	90.1	166.0	172.4
F12_S03	fixed_summary	mid	P1	0	—	—	2190.5	2234.0	2237.6
F12_S03	fixed_summary	mid	P2	0	—	—	877.1	918.5	922.6
F12_S03	fixed_summary	mid	P3	0	—	—	90.8	162.8	169.3
F12_S03	fixed_summary	high	P1	0	—	—	2024.1	2062.5	2066.0
F12_S03	fixed_summary	high	P2	0	—	—	748.8	787.7	790.7
F12_S03	fixed_summary	high	P3	0	—	—	89.3	160.2	166.3
F12_S03	fixed_full	low	P1	90	0.0000	0.0000	1963.8	2094.5	2097.1
F12_S03	fixed_full	low	P2	90	0.0000	0.0000	864.3	926.3	947.9
F12_S03	fixed_full	low	P3	90	0.0000	0.0000	78.2	147.4	153.9
F12_S03	fixed_full	mid	P1	90	0.0000	0.0000	1903.6	1975.4	1976.5
F12_S03	fixed_full	mid	P2	90	0.0000	0.0000	752.0	824.8	829.8
F12_S03	fixed_full	mid	P3	90	0.0000	0.0000	80.7	141.3	147.6
F12_S03	fixed_full	high	P1	90	0.9667	0.0000	2006.1	2050.0	2098.4
F12_S03	fixed_full	high	P2	90	0.9333	0.0000	847.8	892.2	1004.3
F12_S03	fixed_full	high	P3	90	1.0000	1.0000	86.9	150.9	156.0
F12_S03	independent_dpp	low	P1	22	0.0000	0.0000	1937.5	2023.7	2025.5
F12_S03	independent_dpp	low	P2	22	0.0000	0.0000	755.3	832.0	841.6
F12_S03	independent_dpp	low	P3	22	0.0000	0.0000	82.0	151.6	158.6
F12_S03	independent_dpp	mid	P1	45	0.0000	0.0000	7335.7	7390.7	7395.4
F12_S03	independent_dpp	mid	P2	45	0.0000	0.0000	6236.0	6290.3	6295.7
F12_S03	independent_dpp	mid	P3	45	0.0000	0.0000	4851.8	4918.9	4924.2
F12_S03	independent_dpp	high	P1	90	0.9889	0.0000	1867.0	2014.7	2018.8
F12_S03	independent_dpp	high	P2	90	0.9556	0.0000	699.5	814.5	817.0
F12_S03	independent_dpp	high	P3	90	1.0000	1.0000	74.3	134.1	139.7
F12_S03	central_dpp	low	P1	0	—	—	1984.5	2024.7	2030.2
F12_S03	central_dpp	low	P2	0	—	—	838.4	875.1	878.3
F12_S03	central_dpp	low	P3	0	—	—	90.1	161.4	168.3
F12_S03	central_dpp	mid	P1	30	0.8667	0.0000	1949.1	2028.9	2031.1
F12_S03	central_dpp	mid	P2	30	0.7667	0.0000	915.3	954.2	957.1
F12_S03	central_dpp	mid	P3	30	1.0000	1.0000	81.0	147.8	154.9
F12_S03	central_dpp	high	P1	90	0.9667	0.0000	1963.1	2023.1	2025.9
F12_S03	central_dpp	high	P2	90	1.0000	0.0000	856.1	892.5	896.6
F12_S03	central_dpp	high	P3	90	1.0000	1.0000	73.2	133.8	140.1
F12_S03	shielded_dpp	low	P1	0	—	—	2074.6	2116.7	2120.3
F12_S03	shielded_dpp	low	P2	0	—	—	627.9	665.5	668.1
F12_S03	shielded_dpp	low	P3	0	—	—	84.1	154.8	160.5
F12_S03	shielded_dpp	mid	P1	30	0.8000	0.0000	2003.3	2147.9	2150.1

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S03	shielded_dpp	mid	P2	30	0.8333	0.0000	758.1	862.1	892.8
F12_S03	shielded_dpp	mid	P3	30	1.0000	1.0000	86.6	151.2	156.3
F12_S03	shielded_dpp	high	P1	90	0.9333	0.0000	2057.5	2130.4	2139.2
F12_S03	shielded_dpp	high	P2	90	0.9778	0.0000	939.6	989.1	992.6
F12_S03	shielded_dpp	high	P3	90	1.0000	1.0000	77.9	142.0	146.9
F12_S04	fixed_local	low	P1	0	—	—	1879.6	1975.4	1977.9
F12_S04	fixed_local	low	P2	0	—	—	657.9	744.4	746.8
F12_S04	fixed_local	low	P3	0	—	—	91.4	163.7	170.4
F12_S04	fixed_local	mid	P1	0	—	—	1944.5	2017.8	2022.3
F12_S04	fixed_local	mid	P2	0	—	—	724.9	795.4	798.7
F12_S04	fixed_local	mid	P3	0	—	—	95.3	169.4	176.3
F12_S04	fixed_local	high	P1	0	—	—	1896.6	1934.6	1938.1
F12_S04	fixed_local	high	P2	0	—	—	743.8	780.2	783.8
F12_S04	fixed_local	high	P3	0	—	—	87.9	160.4	166.0
F12_S04	fixed_summary	low	P1	0	—	—	1958.8	2049.4	2059.2
F12_S04	fixed_summary	low	P2	0	—	—	719.0	800.6	806.0
F12_S04	fixed_summary	low	P3	0	—	—	87.6	156.2	162.2
F12_S04	fixed_summary	mid	P1	0	—	—	1929.8	2022.4	2027.0
F12_S04	fixed_summary	mid	P2	0	—	—	706.0	830.5	834.7
F12_S04	fixed_summary	mid	P3	0	—	—	89.1	158.3	164.2
F12_S04	fixed_summary	high	P1	0	—	—	1973.7	2015.9	2019.2
F12_S04	fixed_summary	high	P2	0	—	—	595.9	629.4	630.5
F12_S04	fixed_summary	high	P3	0	—	—	88.2	155.6	161.3
F12_S04	fixed_full	low	P1	90	0.0000	0.0000	2135.3	2299.8	2301.8
F12_S04	fixed_full	low	P2	90	0.0000	0.0000	843.4	1033.6	1036.2
F12_S04	fixed_full	low	P3	90	0.0000	0.0000	78.4	140.7	146.6
F12_S04	fixed_full	mid	P1	90	0.0000	0.0000	1990.7	2057.3	2060.3
F12_S04	fixed_full	mid	P2	90	0.0000	0.0000	864.7	897.3	900.8
F12_S04	fixed_full	mid	P3	90	0.0000	0.0000	78.5	141.6	146.8
F12_S04	fixed_full	high	P1	90	0.9556	0.0000	1968.2	2168.1	2172.8
F12_S04	fixed_full	high	P2	90	0.9222	0.0000	746.4	946.3	950.5
F12_S04	fixed_full	high	P3	90	1.0000	1.0000	78.6	142.2	146.8
F12_S04	independent_dpp	low	P1	22	0.0000	0.0000	2113.4	2196.6	2199.2
F12_S04	independent_dpp	low	P2	22	0.0000	0.0000	1004.2	1036.6	1038.8
F12_S04	independent_dpp	low	P3	22	0.0000	0.0000	85.1	154.4	159.8
F12_S04	independent_dpp	mid	P1	45	0.0000	0.0000	1990.7	2228.1	2232.6
F12_S04	independent_dpp	mid	P2	45	0.0000	0.0000	837.2	975.7	977.9
F12_S04	independent_dpp	mid	P3	45	0.0000	0.0000	77.9	143.7	149.7
F12_S04	independent_dpp	high	P1	90	0.9444	0.0000	2007.0	2091.5	2094.1
F12_S04	independent_dpp	high	P2	90	0.9556	0.0000	806.0	927.9	929.9
F12_S04	independent_dpp	high	P3	90	1.0000	1.0000	80.4	145.8	151.8
F12_S04	central_dpp	low	P1	0	—	—	1985.7	2026.6	2029.4
F12_S04	central_dpp	low	P2	0	—	—	669.0	706.4	709.7
F12_S04	central_dpp	low	P3	0	—	—	85.8	158.2	163.9
F12_S04	central_dpp	mid	P1	30	0.6667	0.0000	1908.9	1971.8	1982.7
F12_S04	central_dpp	mid	P2	30	0.7667	0.0000	747.3	842.3	846.1
F12_S04	central_dpp	mid	P3	30	1.0000	1.0000	86.9	156.1	162.2
F12_S04	central_dpp	high	P1	90	0.9556	0.0000	2089.1	2217.9	2220.9
F12_S04	central_dpp	high	P2	90	0.9889	0.0000	855.7	922.1	923.9
F12_S04	central_dpp	high	P3	90	1.0000	1.0000	74.0	133.8	139.0
F12_S04	shielded_dpp	low	P1	0	—	—	1873.0	1929.6	1934.2
F12_S04	shielded_dpp	low	P2	0	—	—	648.1	713.7	716.7
F12_S04	shielded_dpp	low	P3	0	—	—	85.0	156.3	162.7
F12_S04	shielded_dpp	mid	P1	30	0.7667	0.0000	1978.4	2044.0	2047.9
F12_S04	shielded_dpp	mid	P2	30	0.7667	0.0000	795.4	884.6	887.1
F12_S04	shielded_dpp	mid	P3	30	1.0000	1.0000	81.0	144.6	150.5
F12_S04	shielded_dpp	high	P1	90	0.9778	0.0000	1987.3	2025.7	2029.0
F12_S04	shielded_dpp	high	P2	90	0.9111	0.0000	724.4	854.0	858.5
F12_S04	shielded_dpp	high	P3	90	1.0000	1.0000	78.4	144.4	150.5
F12_S05	fixed_local	low	P1	0	—	—	2011.8	2048.3	2050.5
F12_S05	fixed_local	low	P2	0	—	—	735.0	769.7	771.7
F12_S05	fixed_local	low	P3	0	—	—	88.0	158.3	164.5
F12_S05	fixed_local	mid	P1	0	—	—	1826.1	1896.2	1901.4
F12_S05	fixed_local	mid	P2	0	—	—	665.2	704.7	706.8
F12_S05	fixed_local	mid	P3	0	—	—	86.6	158.2	164.3
F12_S05	fixed_local	high	P1	0	—	—	1917.4	1963.7	1966.4
F12_S05	fixed_local	high	P2	0	—	—	763.8	800.2	802.8
F12_S05	fixed_local	high	P3	0	—	—	90.4	169.1	175.5
F12_S05	fixed_summary	low	P1	0	—	—	1836.5	1895.6	1899.3
F12_S05	fixed_summary	low	P2	0	—	—	624.0	695.4	698.3
F12_S05	fixed_summary	low	P3	0	—	—	88.9	160.6	166.5
F12_S05	fixed_summary	mid	P1	0	—	—	1787.3	1904.6	1908.2
F12_S05	fixed_summary	mid	P2	0	—	—	749.4	864.3	868.8
F12_S05	fixed_summary	mid	P3	0	—	—	91.0	163.6	170.1
F12_S05	fixed_summary	high	P1	0	—	—	1869.3	1931.6	1935.0
F12_S05	fixed_summary	high	P2	0	—	—	712.6	774.4	779.4
F12_S05	fixed_summary	high	P3	0	—	—	86.0	152.7	159.9
F12_S05	fixed_full	low	P1	90	0.0000	0.0000	2017.7	2049.2	2051.7
F12_S05	fixed_full	low	P2	90	0.0000	0.0000	779.2	840.4	844.0
F12_S05	fixed_full	low	P3	90	0.0000	0.0000	76.6	140.7	146.2
F12_S05	fixed_full	mid	P1	90	0.0000	0.0000	1920.6	2001.0	2049.4
F12_S05	fixed_full	mid	P2	90	0.0000	0.0000	746.0	797.5	843.3
F12_S05	fixed_full	mid	P3	90	0.0000	0.0000	78.4	144.6	151.0

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S05	fixed_full	high	P1	90	0.9111	0.0000	1861.7	1922.9	1926.9
F12_S05	fixed_full	high	P2	90	0.9222	0.0000	777.4	829.5	831.5
F12_S05	fixed_full	high	P3	90	1.0000	1.0000	81.1	140.9	146.8
F12_S05	independent_dpp	low	P1	22	0.0000	0.0000	1901.3	1976.4	1981.7
F12_S05	independent_dpp	low	P2	22	0.0000	0.0000	654.1	712.2	715.8
F12_S05	independent_dpp	low	P3	22	0.0000	0.0000	81.8	145.1	151.4
F12_S05	independent_dpp	mid	P1	45	0.0000	0.0000	1841.5	1921.5	1925.4
F12_S05	independent_dpp	mid	P2	45	0.0000	0.0000	675.7	737.0	747.7
F12_S05	independent_dpp	mid	P3	45	0.0000	0.0000	74.9	137.5	142.5
F12_S05	independent_dpp	high	P1	90	0.9778	0.0000	1962.0	2083.4	2087.9
F12_S05	independent_dpp	high	P2	90	0.9778	0.0000	886.6	920.4	923.2
F12_S05	independent_dpp	high	P3	90	1.0000	1.0000	76.8	139.6	145.5
F12_S05	central_dpp	low	P1	0	—	—	1957.7	1999.1	2002.0
F12_S05	central_dpp	low	P2	0	—	—	769.8	811.2	814.7
F12_S05	central_dpp	low	P3	0	—	—	85.3	153.6	160.1
F12_S05	central_dpp	mid	P1	30	1.0000	0.0000	1912.7	1940.1	1942.0
F12_S05	central_dpp	mid	P2	30	0.8000	0.0000	793.2	835.2	837.4
F12_S05	central_dpp	mid	P3	30	1.0000	1.0000	82.5	150.1	155.5
F12_S05	central_dpp	high	P1	90	0.9111	0.0000	1946.2	1994.5	1999.5
F12_S05	central_dpp	high	P2	90	0.9778	0.0000	792.1	861.7	865.5
F12_S05	central_dpp	high	P3	90	1.0000	1.0000	77.4	141.8	146.7
F12_S05	shielded_dpp	low	P1	0	—	—	1893.0	1978.6	1982.3
F12_S05	shielded_dpp	low	P2	0	—	—	673.0	754.2	757.7
F12_S05	shielded_dpp	low	P3	0	—	—	84.9	152.9	159.8
F12_S05	shielded_dpp	mid	P1	30	0.5667	0.0000	1903.8	1973.6	1976.1
F12_S05	shielded_dpp	mid	P2	30	0.8333	0.0000	822.2	890.1	892.5
F12_S05	shielded_dpp	mid	P3	30	1.0000	1.0000	81.1	146.8	152.3
F12_S05	shielded_dpp	high	P1	90	0.9222	0.0000	1914.0	1990.0	1992.6
F12_S05	shielded_dpp	high	P2	90	0.9667	0.0000	701.7	774.2	778.5
F12_S05	shielded_dpp	high	P3	90	1.0000	1.0000	80.0	140.8	147.1
F12_S06	fixed_local	low	P1	0	—	—	1932.5	2004.1	2017.5
F12_S06	fixed_local	low	P2	0	—	—	752.8	824.8	837.3
F12_S06	fixed_local	low	P3	0	—	—	90.0	164.6	171.9
F12_S06	fixed_local	mid	P1	0	—	—	1937.2	2042.4	2059.8
F12_S06	fixed_local	mid	P2	0	—	—	673.6	769.8	782.5
F12_S06	fixed_local	mid	P3	0	—	—	90.2	159.5	166.9
F12_S06	fixed_local	high	P1	0	—	—	1909.3	1956.4	1959.2
F12_S06	fixed_local	high	P2	0	—	—	785.0	826.4	829.8
F12_S06	fixed_local	high	P3	0	—	—	90.5	162.6	169.1
F12_S06	fixed_summary	low	P1	0	—	—	1785.0	1821.6	1825.8
F12_S06	fixed_summary	low	P2	0	—	—	614.8	652.1	654.9
F12_S06	fixed_summary	low	P3	0	—	—	88.8	157.6	164.1
F12_S06	fixed_summary	mid	P1	0	—	—	1889.4	1925.6	1929.1
F12_S06	fixed_summary	mid	P2	0	—	—	741.5	779.2	781.7
F12_S06	fixed_summary	mid	P3	0	—	—	85.5	156.6	163.6
F12_S06	fixed_summary	high	P1	0	—	—	1854.5	1988.9	2023.9
F12_S06	fixed_summary	high	P2	0	—	—	659.0	780.9	817.9
F12_S06	fixed_summary	high	P3	0	—	—	91.7	165.0	171.2
F12_S06	fixed_full	low	P1	90	0.0000	0.0000	2001.6	2059.9	2106.9
F12_S06	fixed_full	low	P2	90	0.0000	0.0000	694.5	809.4	850.0
F12_S06	fixed_full	low	P3	90	0.0000	0.0000	80.2	143.3	148.5
F12_S06	fixed_full	mid	P1	90	0.0000	0.0000	1890.5	2019.8	2023.1
F12_S06	fixed_full	mid	P2	90	0.0000	0.0000	652.4	715.4	735.8
F12_S06	fixed_full	mid	P3	90	0.0000	0.0000	77.1	141.6	146.7
F12_S06	fixed_full	high	P1	90	0.9667	0.0000	1966.0	2015.7	2038.6
F12_S06	fixed_full	high	P2	90	0.9111	0.0000	734.7	797.3	799.7
F12_S06	fixed_full	high	P3	90	1.0000	1.0000	80.4	144.3	150.8
F12_S06	independent_dpp	low	P1	22	0.0000	0.0000	2005.8	2040.9	2043.7
F12_S06	independent_dpp	low	P2	22	0.0000	0.0000	696.2	797.4	800.0
F12_S06	independent_dpp	low	P3	22	0.0000	0.0000	84.9	156.4	161.7
F12_S06	independent_dpp	mid	P1	45	0.0000	0.0000	1969.6	1996.1	2000.6
F12_S06	independent_dpp	mid	P2	45	0.0000	0.0000	903.6	932.0	934.8
F12_S06	independent_dpp	mid	P3	45	0.0000	0.0000	79.1	144.5	150.1
F12_S06	independent_dpp	high	P1	90	0.9222	0.0000	1912.8	2042.7	2046.9
F12_S06	independent_dpp	high	P2	90	0.9444	0.0000	730.8	856.4	859.5
F12_S06	independent_dpp	high	P3	90	1.0000	1.0000	73.9	132.8	138.6
F12_S06	central_dpp	low	P1	0	—	—	1945.7	2067.9	2070.5
F12_S06	central_dpp	low	P2	0	—	—	746.1	825.0	827.8
F12_S06	central_dpp	low	P3	0	—	—	84.7	151.8	158.0
F12_S06	central_dpp	mid	P1	30	0.9000	0.0000	2001.6	2059.9	2070.6
F12_S06	central_dpp	mid	P2	30	0.8667	0.0000	869.7	935.4	937.7
F12_S06	central_dpp	mid	P3	30	1.0000	1.0000	89.2	151.7	158.0
F12_S06	central_dpp	high	P1	90	0.9444	0.0000	2175.3	2214.4	2217.3
F12_S06	central_dpp	high	P2	90	0.9667	0.0000	1059.9	1102.3	1105.7
F12_S06	central_dpp	high	P3	90	1.0000	1.0000	79.4	154.0	160.2
F12_S06	shielded_dpp	low	P1	0	—	—	1933.6	1972.0	1976.6
F12_S06	shielded_dpp	low	P2	0	—	—	775.1	817.3	820.9
F12_S06	shielded_dpp	low	P3	0	—	—	89.9	160.3	166.5
F12_S06	shielded_dpp	mid	P1	30	0.8333	0.0000	1830.6	1960.8	1963.5
F12_S06	shielded_dpp	mid	P2	30	0.9000	0.0000	688.1	761.8	764.0
F12_S06	shielded_dpp	mid	P3	30	1.0000	1.0000	82.1	150.9	156.6
F12_S06	shielded_dpp	high	P1	90	0.9778	0.0000	1922.9	1970.4	1972.8
F12_S06	shielded_dpp	high	P2	90	0.9556	0.0000	698.8	740.1	744.4

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S06	shielded_dpp	high	P3	90	1.0000	1.0000	88.2	149.6	155.1
F12_S07	fixed_local	low	P1	0	—	—	1944.9	2011.2	2014.7
F12_S07	fixed_local	low	P2	0	—	—	730.1	809.6	812.8
F12_S07	fixed_local	low	P3	0	—	—	88.8	160.6	167.3
F12_S07	fixed_local	mid	P1	0	—	—	1868.9	1943.8	1948.2
F12_S07	fixed_local	mid	P2	0	—	—	629.0	699.1	702.5
F12_S07	fixed_local	mid	P3	0	—	—	93.9	164.7	171.4
F12_S07	fixed_local	high	P1	0	—	—	2100.5	2245.8	2249.1
F12_S07	fixed_local	high	P2	0	—	—	895.1	1020.1	1024.0
F12_S07	fixed_local	high	P3	0	—	—	88.5	159.7	166.7
F12_S07	fixed_summary	low	P1	0	—	—	1997.9	2132.3	2136.3
F12_S07	fixed_summary	low	P2	0	—	—	675.1	801.3	804.5
F12_S07	fixed_summary	low	P3	0	—	—	84.1	151.1	156.9
F12_S07	fixed_summary	mid	P1	0	—	—	1927.0	1978.8	1985.2
F12_S07	fixed_summary	mid	P2	0	—	—	612.5	680.9	684.0
F12_S07	fixed_summary	mid	P3	0	—	—	87.5	157.8	165.0
F12_S07	fixed_summary	high	P1	0	—	—	2181.8	2223.8	2227.5
F12_S07	fixed_summary	high	P2	0	—	—	613.1	751.0	755.5
F12_S07	fixed_summary	high	P3	0	—	—	84.2	154.1	161.1
F12_S07	fixed_full	low	P1	90	0.0000	0.0000	1825.8	1945.2	1947.3
F12_S07	fixed_full	low	P2	90	0.0000	0.0000	557.2	701.7	705.8
F12_S07	fixed_full	low	P3	90	0.0000	0.0000	81.1	143.6	149.2
F12_S07	fixed_full	mid	P1	90	0.0000	0.0000	2023.6	2274.3	2277.8
F12_S07	fixed_full	mid	P2	90	0.0000	0.0000	771.6	952.8	955.2
F12_S07	fixed_full	mid	P3	90	0.0000	0.0000	77.4	139.8	145.4
F12_S07	fixed_full	high	P1	90	0.9778	0.0000	1983.4	2025.4	2028.5
F12_S07	fixed_full	high	P2	90	0.9333	0.0000	793.0	850.9	872.5
F12_S07	fixed_full	high	P3	90	1.0000	1.0000	79.3	145.9	150.9
F12_S07	independent_dpp	low	P1	22	0.0000	0.0000	1909.6	2101.8	2103.8
F12_S07	independent_dpp	low	P2	22	0.0000	0.0000	717.8	874.3	876.8
F12_S07	independent_dpp	low	P3	22	0.0000	0.0000	83.9	149.3	155.4
F12_S07	independent_dpp	mid	P1	45	0.0000	0.0000	2049.3	2104.3	2107.5
F12_S07	independent_dpp	mid	P2	45	0.0000	0.0000	941.2	966.4	970.3
F12_S07	independent_dpp	mid	P3	45	0.0000	0.0000	85.9	151.4	156.4
F12_S07	independent_dpp	high	P1	90	0.9667	0.0000	1967.2	2170.5	2176.7
F12_S07	independent_dpp	high	P2	90	0.8889	0.0000	782.6	931.8	1045.7
F12_S07	independent_dpp	high	P3	90	0.9889	0.9889	81.2	328.2	333.1
F12_S07	central_dpp	low	P1	0	—	—	2036.6	2077.1	2082.3
F12_S07	central_dpp	low	P2	0	—	—	723.5	764.1	768.0
F12_S07	central_dpp	low	P3	0	—	—	88.3	156.0	161.8
F12_S07	central_dpp	mid	P1	30	0.8667	0.0000	2137.4	2178.3	2188.0
F12_S07	central_dpp	mid	P2	30	0.8667	0.0000	861.1	927.6	931.2
F12_S07	central_dpp	mid	P3	30	1.0000	1.0000	87.2	159.0	164.3
F12_S07	central_dpp	high	P1	90	0.9778	0.0000	2061.6	2089.0	2093.5
F12_S07	central_dpp	high	P2	90	0.9333	0.0000	864.5	941.0	944.8
F12_S07	central_dpp	high	P3	90	1.0000	1.0000	81.4	144.8	150.1
F12_S07	shielded_dpp	low	P1	0	—	—	1894.6	1993.4	1996.3
F12_S07	shielded_dpp	low	P2	0	—	—	628.4	734.7	738.1
F12_S07	shielded_dpp	low	P3	0	—	—	85.6	158.3	163.7
F12_S07	shielded_dpp	mid	P1	30	0.9333	0.0000	2068.2	2134.6	2137.8
F12_S07	shielded_dpp	mid	P2	30	0.9333	0.0000	716.9	752.4	755.3
F12_S07	shielded_dpp	mid	P3	30	1.0000	1.0000	79.0	149.3	154.3
F12_S07	shielded_dpp	high	P1	90	0.9889	0.0000	1891.0	1956.7	1959.9
F12_S07	shielded_dpp	high	P2	90	0.9556	0.0000	704.9	781.2	833.6
F12_S07	shielded_dpp	high	P3	90	1.0000	1.0000	79.4	144.9	151.5
F12_S08	fixed_local	low	P1	0	—	—	1957.4	2013.6	2016.1
F12_S08	fixed_local	low	P2	0	—	—	685.9	732.8	735.9
F12_S08	fixed_local	low	P3	0	—	—	87.0	158.9	165.4
F12_S08	fixed_local	mid	P1	0	—	—	1817.2	1937.7	1954.1
F12_S08	fixed_local	mid	P2	0	—	—	614.7	724.4	744.3
F12_S08	fixed_local	mid	P3	0	—	—	88.0	158.5	164.9
F12_S08	fixed_local	high	P1	0	—	—	1937.6	2017.9	2035.3
F12_S08	fixed_local	high	P2	0	—	—	683.8	776.3	784.7
F12_S08	fixed_local	high	P3	0	—	—	89.9	164.2	170.0
F12_S08	fixed_summary	low	P1	0	—	—	1999.1	2079.8	2083.7
F12_S08	fixed_summary	low	P2	0	—	—	735.8	778.9	786.1
F12_S08	fixed_summary	low	P3	0	—	—	88.0	158.4	164.5
F12_S08	fixed_summary	mid	P1	0	—	—	1992.5	2077.9	2082.7
F12_S08	fixed_summary	mid	P2	0	—	—	745.7	848.4	854.3
F12_S08	fixed_summary	mid	P3	0	—	—	89.3	158.2	164.7
F12_S08	fixed_summary	high	P1	0	—	—	2038.2	2116.8	2119.5
F12_S08	fixed_summary	high	P2	0	—	—	750.9	807.1	811.8
F12_S08	fixed_summary	high	P3	0	—	—	87.3	161.2	167.8
F12_S08	fixed_full	low	P1	90	0.0000	0.0000	2158.3	2203.4	2206.0
F12_S08	fixed_full	low	P2	90	0.0000	0.0000	978.1	1030.7	1034.2
F12_S08	fixed_full	low	P3	90	0.0000	0.0000	75.1	136.5	142.3
F12_S08	fixed_full	mid	P1	90	0.0000	0.0000	1956.9	2041.8	2044.2
F12_S08	fixed_full	mid	P2	90	0.0000	0.0000	904.9	1014.7	1018.5
F12_S08	fixed_full	mid	P3	90	0.0000	0.0000	80.4	142.8	147.6
F12_S08	fixed_full	high	P1	90	0.9333	0.0000	2053.0	2190.7	2193.1
F12_S08	fixed_full	high	P2	90	0.9556	0.0000	908.0	1033.1	1036.6
F12_S08	fixed_full	high	P3	90	1.0000	1.0000	80.5	142.0	146.8
F12_S08	independent_dpp	low	P1	22	0.0000	0.0000	1874.7	2010.2	2012.1

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S08	independent_dpp	low	P2	22	0.0000	0.0000	773.6	889.6	892.9
F12_S08	independent_dpp	low	P3	22	0.0000	0.0000	82.8	150.6	158.3
F12_S08	independent_dpp	mid	P1	45	0.0000	0.0000	1854.8	1911.5	1913.6
F12_S08	independent_dpp	mid	P2	45	0.0000	0.0000	664.3	721.6	724.3
F12_S08	independent_dpp	mid	P3	45	0.0000	0.0000	78.2	140.8	145.7
F12_S08	independent_dpp	high	P1	90	0.9222	0.0000	2146.9	2229.6	2231.4
F12_S08	independent_dpp	high	P2	90	0.9333	0.0000	995.3	1104.8	1107.4
F12_S08	independent_dpp	high	P3	90	1.0000	1.0000	75.3	141.2	146.0
F12_S08	central_dpp	low	P1	0	—	—	2023.5	2362.2	2365.6
F12_S08	central_dpp	low	P2	0	—	—	750.6	1083.7	1087.1
F12_S08	central_dpp	low	P3	0	—	—	83.0	154.3	161.6
F12_S08	central_dpp	mid	P1	30	0.9667	0.0000	2057.4	2105.0	2107.2
F12_S08	central_dpp	mid	P2	30	0.6667	0.0000	784.2	856.1	858.2
F12_S08	central_dpp	mid	P3	30	1.0000	1.0000	79.3	146.0	152.4
F12_S08	central_dpp	high	P1	90	0.9222	0.0000	2048.7	2112.0	2115.0
F12_S08	central_dpp	high	P2	90	0.9556	0.0000	814.7	873.6	877.0
F12_S08	central_dpp	high	P3	90	1.0000	1.0000	80.4	143.1	148.8
F12_S08	shielded_dpp	low	P1	0	—	—	1979.4	2072.9	2075.3
F12_S08	shielded_dpp	low	P2	0	—	—	746.4	815.6	819.0
F12_S08	shielded_dpp	low	P3	0	—	—	87.2	154.6	160.7
F12_S08	shielded_dpp	mid	P1	30	0.7333	0.0000	1895.4	1977.8	1980.2
F12_S08	shielded_dpp	mid	P2	30	0.7000	0.0000	754.9	834.4	837.4
F12_S08	shielded_dpp	mid	P3	30	1.0000	1.0000	84.4	149.1	155.6
F12_S08	shielded_dpp	high	P1	90	0.9556	0.0000	2070.2	2145.7	2150.4
F12_S08	shielded_dpp	high	P2	90	0.9556	0.0000	946.3	1008.9	1011.7
F12_S08	shielded_dpp	high	P3	90	1.0000	1.0000	72.2	135.7	140.7
F12_S09	fixed_local	low	P1	0	—	—	2062.2	2148.9	2153.1
F12_S09	fixed_local	low	P2	0	—	—	835.3	908.6	912.9
F12_S09	fixed_local	low	P3	0	—	—	91.8	166.5	174.5
F12_S09	fixed_local	mid	P1	0	—	—	2017.6	2068.9	2073.1
F12_S09	fixed_local	mid	P2	0	—	—	734.4	788.0	793.2
F12_S09	fixed_local	mid	P3	0	—	—	84.3	154.2	160.5
F12_S09	fixed_local	high	P1	0	—	—	2105.4	2162.6	2166.2
F12_S09	fixed_local	high	P2	0	—	—	862.7	933.5	937.1
F12_S09	fixed_local	high	P3	0	—	—	90.6	162.1	167.7
F12_S09	fixed_summary	low	P1	0	—	—	2035.6	2133.6	2137.1
F12_S09	fixed_summary	low	P2	0	—	—	740.9	828.5	832.8
F12_S09	fixed_summary	low	P3	0	—	—	91.4	166.4	172.2
F12_S09	fixed_summary	mid	P1	0	—	—	2067.9	2207.0	2209.7
F12_S09	fixed_summary	mid	P2	0	—	—	731.5	865.4	869.0
F12_S09	fixed_summary	mid	P3	0	—	—	89.9	163.7	169.8
F12_S09	fixed_summary	high	P1	0	—	—	1931.7	2076.8	2079.4
F12_S09	fixed_summary	high	P2	0	—	—	656.3	775.4	778.8
F12_S09	fixed_summary	high	P3	0	—	—	84.7	152.8	158.7
F12_S09	fixed_full	low	P1	90	0.0000	0.0000	2058.8	2117.3	2124.4
F12_S09	fixed_full	low	P2	90	0.0000	0.0000	756.7	835.5	843.0
F12_S09	fixed_full	low	P3	90	0.0000	0.0000	77.2	141.2	148.3
F12_S09	fixed_full	mid	P1	90	0.0000	0.0000	2005.3	2062.5	2089.6
F12_S09	fixed_full	mid	P2	90	0.0000	0.0000	877.8	931.8	935.3
F12_S09	fixed_full	mid	P3	90	0.0000	0.0000	77.9	136.7	142.7
F12_S09	fixed_full	high	P1	90	0.9111	0.0000	2041.2	2101.3	2106.6
F12_S09	fixed_full	high	P2	90	0.9556	0.0000	795.5	870.5	874.5
F12_S09	fixed_full	high	P3	90	1.0000	1.0000	77.1	141.8	147.0
F12_S09	independent_dpp	low	P1	22	0.0000	0.0000	2044.8	2140.4	2142.8
F12_S09	independent_dpp	low	P2	22	0.0000	0.0000	886.4	937.7	959.9
F12_S09	independent_dpp	low	P3	22	0.0000	0.0000	85.2	148.8	157.8
F12_S09	independent_dpp	mid	P1	45	0.0000	0.0000	2163.0	2195.9	2197.6
F12_S09	independent_dpp	mid	P2	45	0.0000	0.0000	1015.9	1045.1	1065.2
F12_S09	independent_dpp	mid	P3	45	0.0000	0.0000	78.9	140.5	145.3
F12_S09	independent_dpp	high	P1	90	0.9333	0.0000	2125.5	2172.7	2174.9
F12_S09	independent_dpp	high	P2	90	0.9333	0.0000	851.2	955.2	959.5
F12_S09	independent_dpp	high	P3	90	1.0000	1.0000	74.2	135.0	139.5
F12_S09	central_dpp	low	P1	0	—	—	1974.6	2062.7	2073.2
F12_S09	central_dpp	low	P2	0	—	—	700.8	778.4	783.7
F12_S09	central_dpp	low	P3	0	—	—	88.6	160.8	166.0
F12_S09	central_dpp	mid	P1	30	0.9000	0.0000	2105.6	2140.7	2143.5
F12_S09	central_dpp	mid	P2	30	0.7667	0.0000	957.1	996.2	1018.1
F12_S09	central_dpp	mid	P3	30	1.0000	1.0000	85.9	153.4	158.5
F12_S09	central_dpp	high	P1	90	0.9556	0.0000	2328.7	2368.8	2373.6
F12_S09	central_dpp	high	P2	90	0.9778	0.0000	997.1	1036.6	1039.8
F12_S09	central_dpp	high	P3	90	1.0000	1.0000	68.1	135.1	142.1
F12_S09	shielded_dpp	low	P1	0	—	—	1968.5	2040.6	2046.2
F12_S09	shielded_dpp	low	P2	0	—	—	790.4	868.6	870.5
F12_S09	shielded_dpp	low	P3	0	—	—	92.7	162.5	168.8
F12_S09	shielded_dpp	mid	P1	30	0.8667	0.0000	2063.5	2130.1	2140.9
F12_S09	shielded_dpp	mid	P2	30	0.9000	0.0000	834.2	885.0	888.3
F12_S09	shielded_dpp	mid	P3	30	1.0000	1.0000	80.4	145.5	150.6
F12_S09	shielded_dpp	high	P1	90	0.9444	0.0000	2087.2	2138.8	2140.9
F12_S09	shielded_dpp	high	P2	90	0.9444	0.0000	824.1	897.0	899.7
F12_S09	shielded_dpp	high	P3	90	1.0000	1.0000	73.9	137.8	143.0
F12_S10	fixed_local	low	P1	0	—	—	1963.5	2064.8	2068.2
F12_S10	fixed_local	low	P2	0	—	—	761.3	837.0	839.8
F12_S10	fixed_local	low	P3	0	—	—	91.5	166.4	173.2

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S10	fixed_local	mid	P1	0	–	–	2011.0	2050.3	2053.7
F12_S10	fixed_local	mid	P2	0	–	–	715.1	755.4	758.7
F12_S10	fixed_local	mid	P3	0	–	–	90.9	162.7	169.3
F12_S10	fixed_local	high	P1	0	–	–	1937.4	2000.6	2003.6
F12_S10	fixed_local	high	P2	0	–	–	668.6	723.4	726.7
F12_S10	fixed_local	high	P3	0	–	–	93.9	163.6	170.2
F12_S10	fixed_summary	low	P1	0	–	–	2013.5	2103.8	2108.3
F12_S10	fixed_summary	low	P2	0	–	–	823.8	913.5	917.0
F12_S10	fixed_summary	low	P3	0	–	–	84.6	153.6	160.5
F12_S10	fixed_summary	mid	P1	0	–	–	2057.6	2136.9	2140.5
F12_S10	fixed_summary	mid	P2	0	–	–	946.8	1014.3	1018.1
F12_S10	fixed_summary	mid	P3	0	–	–	84.4	155.9	163.6
F12_S10	fixed_summary	high	P1	0	–	–	1943.4	2129.7	2134.3
F12_S10	fixed_summary	high	P2	0	–	–	764.8	939.7	943.1
F12_S10	fixed_summary	high	P3	0	–	–	87.3	163.1	170.0
F12_S10	fixed_full	low	P1	90	0.0000	0.0000	1967.7	2037.9	2041.3
F12_S10	fixed_full	low	P2	90	0.0000	0.0000	858.5	951.5	953.8
F12_S10	fixed_full	low	P3	90	0.0000	0.0000	76.8	138.7	143.3
F12_S10	fixed_full	mid	P1	90	0.0000	0.0000	2146.3	2221.4	2223.5
F12_S10	fixed_full	mid	P2	90	0.0000	0.0000	847.3	890.1	894.4
F12_S10	fixed_full	mid	P3	90	0.0000	0.0000	74.8	132.3	136.9
F12_S10	fixed_full	high	P1	90	0.9778	0.0000	2165.6	2199.9	2204.4
F12_S10	fixed_full	high	P2	90	0.9667	0.0000	871.9	912.1	916.5
F12_S10	fixed_full	high	P3	90	1.0000	1.0000	81.3	142.9	147.6
F12_S10	independent_dpp	low	P1	22	0.0000	0.0000	1921.7	2011.9	2016.7
F12_S10	independent_dpp	low	P2	22	0.0000	0.0000	653.8	788.2	791.3
F12_S10	independent_dpp	low	P3	22	0.0000	0.0000	82.9	150.7	156.6
F12_S10	independent_dpp	mid	P1	45	0.0000	0.0000	1887.2	1966.4	1968.3
F12_S10	independent_dpp	mid	P2	45	0.0000	0.0000	773.0	838.0	840.0
F12_S10	independent_dpp	mid	P3	45	0.0000	0.0000	75.8	141.7	147.3
F12_S10	independent_dpp	high	P1	90	0.8556	0.0000	1965.1	2090.0	2093.2
F12_S10	independent_dpp	high	P2	90	0.9667	0.0000	882.8	981.7	984.1
F12_S10	independent_dpp	high	P3	90	1.0000	1.0000	76.0	142.0	147.0
F12_S10	central_dpp	low	P1	0	–	–	2041.7	2115.9	2119.5
F12_S10	central_dpp	low	P2	0	–	–	834.2	905.3	909.7
F12_S10	central_dpp	low	P3	0	–	–	89.3	164.3	169.0
F12_S10	central_dpp	mid	P1	30	0.9667	0.0000	2076.9	2105.3	2151.2
F12_S10	central_dpp	mid	P2	30	0.9000	0.0000	850.4	929.6	931.8
F12_S10	central_dpp	mid	P3	30	1.0000	1.0000	80.3	148.5	154.8
F12_S10	central_dpp	high	P1	90	0.9556	0.0000	2040.7	2102.1	2106.1
F12_S10	central_dpp	high	P2	90	0.9667	0.0000	905.9	940.3	944.7
F12_S10	central_dpp	high	P3	90	1.0000	1.0000	76.4	140.4	146.7
F12_S10	shielded_dpp	low	P1	0	–	–	2085.7	2128.5	2130.9
F12_S10	shielded_dpp	low	P2	0	–	–	938.2	977.5	981.4
F12_S10	shielded_dpp	low	P3	0	–	–	89.8	160.9	166.9
F12_S10	shielded_dpp	mid	P1	30	0.8000	0.0000	1980.5	2053.7	2058.1
F12_S10	shielded_dpp	mid	P2	30	0.7667	0.0000	729.0	827.8	830.2
F12_S10	shielded_dpp	mid	P3	30	1.0000	1.0000	86.9	154.8	160.6
F12_S10	shielded_dpp	high	P1	90	0.9222	0.0000	2020.4	2096.1	2123.6
F12_S10	shielded_dpp	high	P2	90	0.9778	0.0000	879.9	928.6	932.5
F12_S10	shielded_dpp	high	P3	90	1.0000	1.0000	80.1	143.5	149.0
F12_S11	fixed_local	low	P1	0	–	–	1923.5	1986.9	2001.2
F12_S11	fixed_local	low	P2	0	–	–	743.9	820.1	835.6
F12_S11	fixed_local	low	P3	0	–	–	89.1	161.4	168.1
F12_S11	fixed_local	mid	P1	0	–	–	2032.4	2109.7	2112.1
F12_S11	fixed_local	mid	P2	0	–	–	748.3	824.6	828.8
F12_S11	fixed_local	mid	P3	0	–	–	93.9	169.5	176.2
F12_S11	fixed_local	high	P1	0	–	–	2031.0	2105.3	2109.9
F12_S11	fixed_local	high	P2	0	–	–	736.1	814.3	818.7
F12_S11	fixed_local	high	P3	0	–	–	90.1	162.4	168.3
F12_S11	fixed_summary	low	P1	0	–	–	1928.4	1983.9	1990.1
F12_S11	fixed_summary	low	P2	0	–	–	639.4	714.9	725.7
F12_S11	fixed_summary	low	P3	0	–	–	82.4	153.2	159.2
F12_S11	fixed_summary	mid	P1	0	–	–	2087.9	2136.3	2140.6
F12_S11	fixed_summary	mid	P2	0	–	–	824.7	887.7	894.9
F12_S11	fixed_summary	mid	P3	0	–	–	86.8	157.3	162.5
F12_S11	fixed_summary	high	P1	0	–	–	1941.6	1992.7	1995.2
F12_S11	fixed_summary	high	P2	0	–	–	648.9	703.8	707.1
F12_S11	fixed_summary	high	P3	0	–	–	86.9	155.8	163.5
F12_S11	fixed_full	low	P1	90	0.0000	0.0000	1942.0	2002.2	2006.4
F12_S11	fixed_full	low	P2	90	0.0000	0.0000	724.1	806.1	808.5
F12_S11	fixed_full	low	P3	90	0.0000	0.0000	85.2	150.6	155.4
F12_S11	fixed_full	mid	P1	90	0.0000	0.0000	2070.3	2149.2	2152.6
F12_S11	fixed_full	mid	P2	90	0.0000	0.0000	938.8	980.4	983.0
F12_S11	fixed_full	mid	P3	90	0.0000	0.0000	79.7	142.6	148.2
F12_S11	fixed_full	high	P1	90	0.9111	0.0000	2113.8	2201.4	2203.3
F12_S11	fixed_full	high	P2	90	0.9333	0.0000	912.5	1038.8	1042.4
F12_S11	fixed_full	high	P3	90	1.0000	1.0000	76.8	135.8	141.6
F12_S11	independent_dpp	low	P1	22	0.0000	0.0000	1986.8	2019.7	2022.3
F12_S11	independent_dpp	low	P2	22	0.0000	0.0000	764.0	802.3	804.8
F12_S11	independent_dpp	low	P3	22	0.0000	0.0000	88.9	156.0	162.7
F12_S11	independent_dpp	mid	P1	45	0.0000	0.0000	2001.6	2087.2	2091.6
F12_S11	independent_dpp	mid	P2	45	0.0000	0.0000	824.6	904.2	906.9

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S11	independent_dpp	mid	P3	45	0.0000	0.0000	78.9	141.5	146.5
F12_S11	independent_dpp	high	P1	90	0.9000	0.0000	2133.6	2182.2	2184.9
F12_S11	independent_dpp	high	P2	90	0.9444	0.0000	981.2	1046.9	1051.6
F12_S11	independent_dpp	high	P3	90	1.0000	1.0000	71.8	129.6	136.5
F12_S11	central_dpp	low	P1	0	—	—	2043.7	2116.0	2120.4
F12_S11	central_dpp	low	P2	0	—	—	783.2	865.3	868.8
F12_S11	central_dpp	low	P3	0	—	—	93.5	162.4	168.8
F12_S11	central_dpp	mid	P1	30	0.8000	0.0000	2106.5	2167.1	2169.4
F12_S11	central_dpp	mid	P2	30	0.8333	0.0000	934.4	991.7	995.4
F12_S11	central_dpp	mid	P3	30	1.0000	1.0000	80.0	146.7	151.1
F12_S11	central_dpp	high	P1	90	0.8889	0.0000	1923.1	1979.1	1982.6
F12_S11	central_dpp	high	P2	90	0.9444	0.0000	689.4	782.6	785.7
F12_S11	central_dpp	high	P3	90	1.0000	1.0000	75.7	135.1	140.8
F12_S11	shielded_dpp	low	P1	0	—	—	2014.5	2077.7	2080.4
F12_S11	shielded_dpp	low	P2	0	—	—	735.4	818.5	822.9
F12_S11	shielded_dpp	low	P3	0	—	—	87.8	158.6	165.2
F12_S11	shielded_dpp	mid	P1	30	0.8333	0.0000	1884.2	1947.6	1950.1
F12_S11	shielded_dpp	mid	P2	30	0.8667	0.0000	802.2	834.0	882.2
F12_S11	shielded_dpp	mid	P3	30	1.0000	1.0000	79.7	142.8	149.4
F12_S11	shielded_dpp	high	P1	90	0.9000	0.0000	1866.9	1926.8	1928.9
F12_S11	shielded_dpp	high	P2	90	0.9333	0.0000	651.9	721.0	724.2
F12_S11	shielded_dpp	high	P3	90	1.0000	1.0000	78.9	143.3	149.7
F12_S12	fixed_local	low	P1	0	—	—	2021.6	2060.8	2064.9
F12_S12	fixed_local	low	P2	0	—	—	851.3	891.4	896.0
F12_S12	fixed_local	low	P3	0	—	—	89.4	162.6	169.8
F12_S12	fixed_local	mid	P1	0	—	—	2065.4	2110.6	2113.4
F12_S12	fixed_local	mid	P2	0	—	—	880.8	918.2	921.2
F12_S12	fixed_local	mid	P3	0	—	—	95.8	167.4	173.3
F12_S12	fixed_local	high	P1	0	—	—	2028.6	2158.6	2162.3
F12_S12	fixed_local	high	P2	0	—	—	775.1	837.5	843.8
F12_S12	fixed_local	high	P3	0	—	—	92.3	164.3	171.5
F12_S12	fixed_summary	low	P1	0	—	—	1910.3	1985.2	1989.2
F12_S12	fixed_summary	low	P2	0	—	—	718.9	806.6	808.8
F12_S12	fixed_summary	low	P3	0	—	—	88.1	159.9	165.1
F12_S12	fixed_summary	mid	P1	0	—	—	1893.9	1959.5	1962.4
F12_S12	fixed_summary	mid	P2	0	—	—	613.9	675.0	678.6
F12_S12	fixed_summary	mid	P3	0	—	—	88.1	155.1	161.6
F12_S12	fixed_summary	high	P1	0	—	—	2217.8	2256.9	2260.5
F12_S12	fixed_summary	high	P2	0	—	—	1057.1	1098.5	1102.7
F12_S12	fixed_summary	high	P3	0	—	—	82.4	150.6	156.6
F12_S12	fixed_full	low	P1	90	0.0000	0.0000	2056.0	2127.3	2128.7
F12_S12	fixed_full	low	P2	90	0.0000	0.0000	831.5	884.5	889.1
F12_S12	fixed_full	low	P3	90	0.0000	0.0000	85.1	157.7	163.5
F12_S12	fixed_full	mid	P1	90	0.0000	0.0000	1859.9	1997.8	2002.3
F12_S12	fixed_full	mid	P2	90	0.0000	0.0000	697.3	841.4	844.7
F12_S12	fixed_full	mid	P3	90	0.0000	0.0000	83.0	145.3	151.3
F12_S12	fixed_full	high	P1	90	0.9222	0.0000	2070.3	2135.0	2139.0
F12_S12	fixed_full	high	P2	90	0.9444	0.0000	956.1	1001.6	1003.5
F12_S12	fixed_full	high	P3	90	1.0000	1.0000	80.0	138.6	145.2
F12_S12	independent_dpp	low	P1	22	0.0000	0.0000	1001.7	1038.7	1041.5
F12_S12	independent_dpp	low	P2	22	0.0000	0.0000	864.1	915.7	918.1
F12_S12	independent_dpp	low	P3	22	0.0000	0.0000	85.0	157.5	162.6
F12_S12	independent_dpp	mid	P1	45	0.0000	0.0000	2161.5	2193.0	2197.7
F12_S12	independent_dpp	mid	P2	45	0.0000	0.0000	860.3	896.0	897.8
F12_S12	independent_dpp	mid	P3	45	0.0000	0.0000	77.0	138.3	143.6
F12_S12	independent_dpp	high	P1	90	0.9444	0.0000	2004.9	2052.8	2056.1
F12_S12	independent_dpp	high	P2	90	0.9444	0.0000	778.7	819.1	822.4
F12_S12	independent_dpp	high	P3	90	1.0000	1.0000	74.8	135.8	141.5
F12_S12	central_dpp	low	P1	0	—	—	2072.6	2369.1	2372.7
F12_S12	central_dpp	low	P2	0	—	—	865.3	1186.7	1191.2
F12_S12	central_dpp	low	P3	0	—	—	86.1	161.1	167.6
F12_S12	central_dpp	mid	P1	30	0.5333	0.0000	1883.9	1935.5	1939.8
F12_S12	central_dpp	mid	P2	30	0.7667	0.0000	659.2	732.8	735.7
F12_S12	central_dpp	mid	P3	30	1.0000	1.0000	83.3	144.6	150.7
F12_S12	central_dpp	high	P1	90	0.9222	0.0000	2095.8	2187.1	2190.5
F12_S12	central_dpp	high	P2	90	0.9444	0.0000	846.9	919.8	922.5
F12_S12	central_dpp	high	P3	90	1.0000	1.0000	81.8	147.6	152.4
F12_S12	shielded_dpp	low	P1	0	—	—	1879.5	1960.7	1964.6
F12_S12	shielded_dpp	low	P2	0	—	—	695.3	756.2	759.7
F12_S12	shielded_dpp	low	P3	0	—	—	87.3	156.0	161.3
F12_S12	shielded_dpp	mid	P1	30	0.9000	0.0000	2041.9	2073.2	2075.5
F12_S12	shielded_dpp	mid	P2	30	0.9333	0.0000	803.7	835.2	837.5
F12_S12	shielded_dpp	mid	P3	30	1.0000	1.0000	83.1	147.1	152.7
F12_S12	shielded_dpp	high	P1	90	0.9444	0.0000	1962.8	2023.5	2026.7
F12_S12	shielded_dpp	high	P2	90	0.9444	0.0000	736.5	774.2	776.1
F12_S12	shielded_dpp	high	P3	90	1.0000	1.0000	74.7	136.9	142.8

VII. Release-Lateness Diagnostic

Independent nominal clocks, the three-probe barrier, unequal first-iteration waiting, relative sleep, and no absolute catch-up are source/log supported. The specific OS/network reason for the recurring proposal-arrival order is UNKNOWN.

TABLE VI: Release-lateness diagnostic from accepted formal timestamps (ms).

Device	Events	p50	p95	p99	Unproven residual
P1	19440	1972.34	2173.36	2326.13	The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN.
P2	19440	763.90	976.78	1080.30	The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN.
P3	19440	83.56	154.08	167.94	The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN.

VIII. Finite- n Scaling Conformance

This is discrete-event conformance, not a 32-probe hardware scalability experiment.

TABLE VII: All finite- n discrete-event conformance cells.

n	ρ	C (B)	k^*	Sim. k	Theory frac.	Sim. frac.	Match
3	1/3	131072	0	0	0.000000	0.000000	PASS
3	1/2	196608	1	1	0.333333	0.333333	PASS
3	2/3	262144	1	1	0.333333	0.333333	PASS
3	3/4	294912	2	2	0.666667	0.666667	PASS
3	1/1	393216	3	3	1.000000	1.000000	PASS
8	1/3	349525	1	1	0.125000	0.125000	PASS
8	1/2	524288	3	3	0.375000	0.375000	PASS
8	2/3	699050	4	4	0.500000	0.500000	PASS
8	3/4	786432	5	5	0.625000	0.625000	PASS
8	1/1	1048576	8	8	1.000000	1.000000	PASS
16	1/3	699050	3	3	0.187500	0.187500	PASS
16	1/2	1048576	6	6	0.375000	0.375000	PASS
16	2/3	1398101	9	9	0.562500	0.562500	PASS
16	3/4	1572864	11	11	0.687500	0.687500	PASS
16	1/1	2097152	16	16	1.000000	1.000000	PASS
32	1/3	1398101	7	7	0.218750	0.218750	PASS
32	1/2	2097152	13	13	0.406250	0.406250	PASS
32	2/3	2796202	19	19	0.593750	0.593750	PASS
32	3/4	3145728	22	22	0.687500	0.687500	PASS
32	1/1	4194304	32	32	1.000000	1.000000	PASS

IX. Accepted and Excluded Attempts

Accepted attempts are S01-a01, S02-a01, S03-a02, S04-a02, S05-a01, S06-a01, S07-a02, and S08–S12-a01. S03-a01, S04-a01, and S07-a01 are excluded infrastructure attempts and are not stitched into accepted sessions.

X. Detector and Replay Limitations

The workload is frozen CICIOT2023-derived feature-vector replay, not packet replay or an uncontrolled deployment. The formal occurrence set excludes Recon-PortScan, and exact feature-vector overlap exists across frozen partitions. Perfect correctness on the formal occurrences is therefore a controlled diagnostic rather than an IDS-superiority result.

XI. Raw Audit and Methodology-Repair Verdicts

The full-raw audit rebuilt 216/216 accepted arms and 58,320 measured events with zero raw-to-metric mismatch and zero shared-cap violation across 63,720 service rounds. Its verdict is `PASS_WITH_ISSUES` because the raw bundle lacks direct artifacts for the S03-a01 SSD incident and the S07 pre-hardware launcher abort. The methodology repair is `PASS` with retained rows `A/B/C/D = 432/240/648/20` and zero failed runs.

XII. Artifact Access

Machine-readable CSV/JSON sources, number provenance, claim audit, build report, validation report, and SHA-256 manifest accompany this supplementary PDF. Third-party dataset redistribution rights remain outside the scope of this package.